RCF-FW-v7.0
RELATIONAL COARSE-GRAINING FRAMEWORK

The RCF Framework, Post-Arc Rewrite

From the Hessian identity to the picture's boundary statement: a single coherent specification subsuming v6.5-Protocol#20, v6.5-RF8, and the six-round AKLT substrate investigation.

Document ID	RCF-FW-v7.0
Status	[Established -- Post-Arc Specification]
Lineage	v6.5-#20 + v6.5-RF8 + v6.1-RF1-audit-patch + six-round AKLT arc (delta / epsilon / zeta / eta / eta-extreme / Option A) + primitive-vs-emergent axiom
Supersedes	All RCF-RES-v6.x increments as canonical framework (they remain as audit-trail artifacts)
Compliance	Protocols #15-#20 enforced; primitive-vs-emergent axiom applied retroactively
Auditor	Z.ai (Super Z collaboration) -- automated verification + crosscheck
Verification	Exact rational arithmetic (sympy); Lanczos eigsh on dim-81 Hilbert space; machine precision confirmed across all six rounds
Date	2026-08-29

Z.ai | Super Z collaboration | post-arc closure + substrate-change launch

Document Header

Field	Value
Document ID	RCF-FW-v7.0
Title	The RCF Framework, Post-Arc Rewrite
Status	[Established -- Post-Arc Specification] Co-equal canonical framework document with the RCF-RES-v6.x audit-trail artifacts (which remain in force as historical record).
Lineage	v6.5-Protocol#20 (Type-Tagging) + v6.5-RF8 (Hessian Minimization) + v6.1-RF1-audit-patch + six-round AKLT substrate arc (delta, epsilon, zeta, eta, eta-extreme, Option A) + primitive-vs-emergent foundational axiom (user's final note)
Supersedes	All RCF-RES-v6.x increments as canonical framework reference. They remain as audit-trail artifacts; their mathematical content is preserved verbatim here, their status vocabulary is inherited unchanged, and their lineage annotations are retained.
Compliance	Protocols #15 (Independent Recomputation), #16 (Variational Principle), #17 (Parity & Extremum Derivative), #18 (Numerical Equivalence Machine-Check), #19 (Bibliographic External Lock), #20 (Type-Tagging of Vanishing Claims) all enforced. Primitive-vs-emergent axiom applied retroactively across the VN register and the AKLT arc.
Auditor	Z.ai (Super Z collaboration) -- automated verification + crosscheck; exact rational arithmetic where possible, Lanczos eigsh on dim-81 AKLT Hilbert space, machine-precision confirmed across all six arc rounds.
Date	2026-08-29

Contents

1. Motivation: From Black-Hole Information to Relational Coarse-Graining

The Relational Coarse-Graining Framework (RCF) was inaugurated to address a single question whose precise formulation has shifted across three decades of black-hole physics: how does information survive the strain that a strong gravitational field imposes on the Hilbert space that describes it? The framework's founders declined the two standard exits from this question. They declined the firewall answer (which makes the survival a non-event by discarding the interior), and they declined the unitarity-by-decoherence answer (which salvages the global unitary but loses the operational meaning of local observables under strain). Both exits avoid the question; RCF attempts to answer it.

The framework's central wager is that the right level of description for this problem is neither the local operator algebra of an effective field theory nor the global state of a closed quantum system, but a *relational coarse-graining* layer in between. A relational coarse-graining is a decomposition of the operator algebra into equivalence classes that respect a designated notion of locality and symmetry, plus the coarse-grained observables that the equivalence classes generate. The framework's wager is that this layer, properly constructed, supports a notion of robustness under strain that neither the local nor the global description can provide.

Across its six-year development the framework has matured through four phases. The first phase (v1-v4) established the operator-algebraic foundations, the verified-number (VN) registry, and the protocol system (#15-#19). The second phase (v6.0-v6.5) introduced the eight-tag status vocabulary, the Type-A / Type-B epistemic classification (Protocol #20), and the Hessian identity foundation (RF-8), discharging postulate S5 and internalizing RF-4. The third phase, which this document closes, is the six-round AKLT substrate arc (delta, epsilon, zeta, eta, eta-extreme, Option A) -- an attempt to realize the projection picture concretely on a controlled substrate. The fourth phase, which this document launches, is the substrate-change fork.

The arc's final verdict is negative on every dynamical form the AKLT substrate could host. This is not a defeat. It is the substrate answering the framework's question in the only way a substrate can -- by saying, at every location it possesses and in every regime the framework can dial, that the projection picture does not survive here. The arc leaves behind four permanent scientific facts (a falsified collapse-ordering, a falsified role-transfer, a characterized conformal floor artifact, a characterized multiplet artifact), a complete map of the substrate's trade-off surface (gap vs. information, monotone, no interior extremum), a measured endpoint (h -> infinity, all channels cloak together), and a diagnostic taxonomy of the artifacts the substrate's instruments produce when their outputs are read naively.

Most importantly, the arc surfaced a foundational correction that this document now absorbs as a fifth axiom: in RCF every physical observable is emergent, not primitive. The AKLT substrate -- like nearly every spin-chain substrate -- is built with primitive observables (the spin operators S^z, S^+, S^-, their correlations, their string and edge composites). The arc treated these as primitive because that is what the substrate hands you. But RCF's foundational commitment is that physical observables are *always* emergent from the relational coarse-graining, never primitives of the framework. The mismatch between the substrate's primitive-observable algebra and RCF's emergent-observable commitment is, in retrospect, the structural source of the negative verdict. The picture's boundary statement in section 8 is the formal expression of this insight.

The framework's structural commitments remain what they were before the arc. The Hessian identity g_FS = (1/2) Hess(Phi) (RF-8) is the foundational Type-A tautology on which the parabolic gradient flow is built. The Type-A / Type-B classification (Protocol #20) remains the mandatory vocabulary for every vanishing claim. The certified master hierarchy (delta_DL <= delta_Knabe^(n) <= Delta_infinity <= epsilon_n) remains the verified-number spine. The protocol system (#15-#20) remains the standing audit apparatus. What changes is the picture's scope: the projection picture survives *only* as the standard-GR cloaking asymmetry, and is falsified in every dynamical form the AKLT substrate could host. The next substrate must be selected by a criterion that the primitive-vs-emergent axiom makes explicit (section 8 and Appendix A).

MOTIVATION VERDICT: The arc closes the AKLT substrate's question with a negative answer at every location and regime. The negative answer is permanent scientific capital. The framework absorbs the primitive-vs-emergent axiom as foundational, scopes the projection picture to its surviving (standard-GR) form, and launches a substrate-change fork whose selection criterion is the axiom's operational consequence.

2. Foundational Axioms

The framework's foundational commitments are organized as five axioms. The first four (S1-S4) are inherited unchanged from v6.0-F; the fifth (S5, the metric-flow well-posedness postulate) was discharged in v6.5-RF8 and is no longer primitive. In its place, this document introduces a new fifth axiom (S5') -- the primitive-vs-emergent axiom -- which the arc surfaced as a structural correction that the framework must absorb going forward. The primitive count remains four (the Hessian identity absorbed the old S5 in v6.5-RF8; the primitive-vs-emergent axiom is a structural axiom, not a primitive postulate, because it constrains the framework's *interpretation* rather than its mathematical content).

2.1 The Four Primitive Postulates (S1-S4, inherited)

S1 (Hilbert space structure). The framework operates on a finite-dimensional complex Hilbert space H. All operator-algebraic constructions take place in B(H), the bounded operators on H. The choice of finite-dimensionality is a regularization: the framework's theorems are first proven in finite dimension and then thermodynamic limits are taken where the limit exists. The Hilbert space carries no primitive observable structure -- spin operators, position operators, etc. are substrate-supplied, not framework-supplied.

S2 (Hamiltonian spectrum). The framework takes as input a Hamiltonian H with spectrum sigma(H) = {E_0, E_1, E_2, ...}. The gap Delta = E_1 - E_0 and the thermodynamic-limit gap Delta_infinity are framework primitives. The certified master hierarchy (section 11) is the verified-number spine on which the framework's gap statements rest. Substrates that fail to produce a computable gap (e.g. gapless critical systems with no finite-size regularization) are out of scope for the framework's robustness analysis.

S3 (Symmetry group and moment map). The framework takes as input a symmetry group G of the Hamiltonian, together with the moment map F_MM: H -> Lie(G)* that specifies the algebraic compatibility of a state with the symmetry structure. The constraint manifold M_constraint = {psi in CP(H) : F_MM(psi) = 0} is the locus on which the parabolic gradient flow of section 6 is defined. This is Type-A [=] by the moment-map construction (the moment-map equation is a symmetry-forced algebraic identity; registered as VN-20).

S4 (Variational principle). The ground state is characterized by the variational principle E_0 = inf_psi <psi|H|psi>. The parabolic gradient flow on (M_constraint, g_FS) is a Type-B mechanism whose fixed points realize the variational characterization. The geometric burden B_geom = Var(g_FS) is bounded below by zero (variance is non-negative), which is the variational structure Protocol #16 requires.

2.2 S5 Discharged (v6.5-RF8)

The v6.0-F framework postulated as primitive S5 that "the metric flow on quantum state space is well-posed." With the Hessian identity g_FS = (1/2) Hess(Phi) (section 5, Theorem 3.1), this is no longer primitive. The Hessian identity provides the Kahler structure on CP(H), the bounded-below functional B_geom = Var(g_FS) automatically generates a well-posed gradient flow on M_constraint, and S5 becomes a derived consequence. The primitive count dropped from 5 to 4 in v6.5-RF8 and remains 4 in this document.

S5 (Old, v6.0-F): The metric flow d_t psi = -grad B_geom is well-posed. [Primitive postulate]
S5 (New, v6.5-RF8 onward): The metric flow is well-posed BECAUSE g_FS = (1/2) Hess(Phi) is the Kahler structure on CP(H), and B_geom is bounded below. [Derived consequence]

2.3 S5' -- The Primitive-vs-Emergent Axiom (NEW, structural)

The arc surfaced a structural correction that this document absorbs as a structural axiom (not a primitive postulate, because it constrains interpretation rather than mathematics). The axiom is given formally in section 3; its operational consequence (the substrate-evaluation criterion) is given in section 8 and Appendix A. The axiom is summarized here for reference:

> S5' (Structural axiom): In RCF, every physical observable is emergent from the relational coarse-graining, never primitive. Substrates whose primitive-observable algebra is too distant from an emergent interpretation are ruled out as hosts for the projection picture.

The axiom's content is interpretive: it tells the framework how to read substrate observables. The AKLT substrate hands the framework spin operators S^z, S^+, S^-, and their correlations (string, edge, bulk) as primitives. The arc treated them as primitives because that is what the substrate hands you. But RCF's commitment is that physical observables are *always* emergent from the relational coarse-graining, never primitives of the framework. The mismatch between the substrate's primitive-observable algebra and RCF's emergent-observable commitment is the structural source of the negative verdict on every dynamical form the AKLT substrate could host. The next substrate must be selected so that its primitive-observable algebra admits an emergent interpretation within RCF -- the selection criterion in Appendix A makes this test explicit.

2.4 Axiom Inventory (post-arc)

Tag	Axiom	Status	Type
S1	Hilbert space structure H (finite-dim, regularized)	[Established]	Primitive
S2	Hamiltonian spectrum sigma(H); gap Delta; Delta_infinity	[Established]	Primitive
S3	Symmetry group G + moment map F_MM; constraint manifold M_constraint = {F_MM = 0}	[Established -- Type-A [=] by moment map]	Primitive
S4	Variational principle E_0 = inf <psi|H|psi>; B_geom bounded below by 0	[Established -- Type-A [=] by variance]	Primitive
S5	Metric-flow well-posedness	DISCHARGED v6.5-RF8 (derived from Hessian identity)	Former primitive
S5'	Primitive-vs-emergent axiom (every physical observable emergent, never primitive)	[Established -- Structural axiom, this document]	Structural (not primitive)

2.5 Standing Protocol Apparatus

The framework's standing audit apparatus (Protocols #15-#20) is inherited unchanged from v6.5. Every claim in this document is compliant with all six protocols. The protocols are summarized here for reference; their full text is in the v6.5-Protocol#20 and v6.5-RF8 documents (audit-trail artifacts).

#	Protocol	Enforcement
15	Independent Recomputation	Symbolic verification via SymPy; exact rational arithmetic where possible.
16	Variational Principle Enforcement	All flows must be variational (B_geom monotone decreasing).
17	Parity & Extremum Derivative	Curvature-level structures parity-respecting at the Kahler level.
18	Numerical Equivalence Machine-Check	Identities verified to exact rational arithmetic (e.g. g_FS / Hess(Phi) = 1/2).
19	Bibliographic External Lock	No new external citation claims; standard results cited by source.
20	Type-Tagging of Vanishing Claims	Every vanishing claim carries an explicit [=] (Type-A) or [->: rate] (Type-B) tag. See section 4.

3. The Primitive-vs-Emergent Distinction

The primitive-vs-emergent distinction is the foundational correction this document absorbs from the AKLT arc. The distinction is structural: it tells the framework how to read the observables a substrate hands it. The AKLT substrate, like nearly every spin-chain substrate, hands the framework a primitive-observable algebra (the spin operators and their correlations). RCF's foundational commitment is that physical observables are *always* emergent from the relational coarse-graining, never primitives of the framework. The mismatch is the structural source of the negative verdict on every dynamical form the AKLT substrate could host.

3.1 The Axiom (Formal Statement)

AXIOM S5' (Primitive-vs-Emergent). In RCF, every physical observable is emergent from the relational coarse-graining C_L: B(H) -> B(H)/~, never primitive. A substrate's primitive-observable algebra A_prim is admissible as a host for the projection picture only if there exists a coarse-graining C_L such that A_prim admits an emergent interpretation -- i.e. every element of A_prim is expressible as a function of C_L-equivalence class observables. The distance |A_prim - A_emergent(C_L)| is the substrate's primitive-emergent gap; the gap must be finite (and ideally zero) for the substrate to host the picture.

The axiom has three layers, which must be read together.

(i) Interpretive layer. The axiom tells the framework how to read substrate observables. A substrate observable O is never a primitive of RCF, even if the substrate's own algebra treats it as primitive. RCF reads O as a coarse-grained emergent from a (possibly different) relational structure on the substrate.

(ii) Admissibility layer. A substrate is admissible as a host for the projection picture only if its primitive-observable algebra admits an emergent interpretation within RCF. The admissibility test is constructive: the framework must exhibit a coarse-graining C_L such that every primitive observable is expressible as a function of C_L-class observables. Substrates that fail the admissibility test (no such C_L exists, or the gap is too large) are ruled out as hosts for the picture -- but not as hosts for the rest of the framework's structural content (Hessian identity, certified hierarchy, etc.).

(iii) Distance layer. The primitive-emergent gap |A_prim - A_emergent(C_L)| measures how distant a substrate is from hosting the picture cleanly. A gap of zero means the substrate's primitives are *identical* to the framework's emergents -- the cleanest possible host. A finite positive gap means the substrate's primitives are expressible as functions of emergents but require nontrivial coarse-graining. An infinite gap means no C_L exists; the substrate is ruled out as a host.

3.2 The AKLT Substrate's Primitive-Obsevable Algebra

The AKLT spin-1 chain (the substrate of the six-round arc) hands the framework the following primitive-observable algebra:

Primitive	Definition	Channel	Used in round
S^z_i	z-component of spin-1 at site i	existence	delta, epsilon, zeta, eta, eta-extreme, Option A
S^+_i, S^-_i	ladder operators at site i	existence (off-diag)	delta (string)
<S^z_i S^z_j>	two-site z-z correlation	distance	delta, epsilon, zeta, Option A (bulk)
<S^z_i S^z_{i+1}>	nearest-neighbor z-z	direction	delta, epsilon, zeta, Option A

Primitive	Definition	Channel	Used in round
O_string(i,j) = prod S^z_k (i≤k≤j)	string order parameter	string/string-nonlocal	delta, epsilon, zeta, Option A
<(S^z_0)^2 + (S^z_{N-1})^2>	edge existence (OBC)	edge existence	Option A
<S^z_0 S^z_{N-1}>	edge-edge correlation (OBC)	edge distance	Option A
<O_string(0, N-1)>	edge string order (OBC)	edge string	Option A
Delta = E_1 - E_0	bulk gap (multiplet-corrected)	gap (R4)	all rounds

Every observable in the table above is primitive in the AKLT substrate's algebra: it is defined directly in terms of spin operators on sites, without reference to any coarse-graining equivalence class. The arc treated them as primitives because that is what the substrate hands you. But RCF's commitment is that every physical observable is emergent from C_L. The substrate's primitive-emergent gap is therefore nonzero, and in fact large: there is no coarse-graining C_L on the AKLT chain that expresses all of S^z, <S^z S^z>, O_string, and the edge observables as emergents of a single relational structure.

3.3 Why the Mismatch Produced the Negative Verdict

The arc's negative verdict on every dynamical form the AKLT substrate could host is, in retrospect, the structural consequence of the primitive-emergent mismatch. Consider what the projection picture requires: a relational structure on the substrate such that, when the substrate is strained, the relational observables survive (R3) and a stable floor is preserved (R4). The picture's projection mechanism is an isomorphism between the substrate's bulk algebra and a boundary algebra; the survival of relational information under strain depends on this isomorphism being nontrivial -- i.e. on the substrate's primitives being emergents of a relational structure that admits a boundary-bulk decomposition.

The AKLT substrate's primitives do not admit such a decomposition cleanly. The spin operators S^z_i are local, site-indexed, and have no natural boundary-bulk split (the edge S^z_0 is the same operator as the bulk S^z_3, just at a different site). The string order O_string(i,j) is nonlocal but is a product of local S^z's, so it inherits the no-boundary-bulk-split property. The edge observables are *defined* as boundary-local versions of bulk observables; they are not emergents of a boundary algebra distinct from the bulk algebra.

The SPT-edge protection at h=0 (Option A measured edge = 2 x bulk) is the cleanest case of boundary structure the AKLT substrate exhibits. But this protection is fragile (h~1 destroys it) and does not survive V_strain. The fragility is the symptom of the primitive-emergent mismatch: the edge observables are *defined* as primitives at the boundary, not as emergents of a boundary algebra that the projection mechanism would produce. When strain destroys the SPT structure, the edge observables collapse alongside the bulk observables because they were never emergents of a distinct boundary algebra in the first place.

3.4 Operational Consequence: The Substrate-Evaluation Criterion

The axiom's operational consequence is a substrate-evaluation criterion that any candidate substrate must pass before being admitted as a host for the projection picture. The criterion is stated formally in Appendix A and applied to the four candidate substrates in the substrate-change fork; it is summarized here.

> CRITERION (Substrate admissibility for the projection picture). A substrate S with primitive-observable algebra A_prim(S) is admissible as a host only if there exists a coarse-graining C_L on S such that (i) every primitive observable in A_prim(S) is expressible as a function of C_L-class observables; (ii) C_L admits a nontrivial boundary-bulk decomposition (boundary algebra not identical to bulk algebra); and (iii) the

boundary-bulk decomposition is kinematic (phase-independent) rather than SPT-dynamical (phase-dependent).

Condition (iii) is the one the AKLT substrate violates: the AKLT SPT-edge protection is phase-dependent (it exists in the Haldane phase, h~0; it is destroyed by strain, h~1). The next substrate must supply a boundary-bulk decomposition that is kinematic -- i.e. built into the substrate's definition rather than into a phase the substrate happens to occupy. The four candidate substrates in Appendix A are evaluated against this criterion.

SECTION 3 VERDICT: The primitive-vs-emergent axiom is [Established -- Structural axiom]. Its operational consequence is the substrate-evaluation criterion (Appendix A). The AKLT substrate's primitive-emergent gap is large and is the structural source of the negative verdict on every dynamical form the substrate could host.

4. The Type-A / Type-B Epistemic Classification

The Type-A / Type-B classification was ratified in v6.5-Protocol#20 to permanently prevent a category error that surfaced during the RF-2 audit cycle. The error was a structural conflation between two mathematically distinct categories of "vanishing" statements: an exact identity f(x) = 0 forced a priori by symmetry / algebra / representation theory (Type-A), versus an asymptotic limit lim_{n->inf} f(n) = 0 whose rate O(n^-p) is determined a posteriori by physical mechanism (Type-B). Both are vanishing claims but they differ in logical force: the Type-A identity constrains the FORM of related quantities a priori; the Type-B limit constrains only the existence of convergence, not the rate. This classification is preserved unchanged in v7.0; all vanishing claims in this document carry explicit [=] (Type-A) or [->: rate] (Type-B) tags.

4.1 Type-A [Exact Identity, tag: [=]]

A claim of the form f(x) = 0, where the vanishing is forced by (a) symmetry (parity, gauge, time-reversal, etc.), or (b) algebraic identity (commutator, Jacobi, C*-algebra identity, Kahler-potential identity), or (c) representation theory (selection rules, Schur's lemma).

Logical force. A Type-A identity constrains the FORM of related quantities a priori. Specifically, a Type-A identity may force another quantity to have a specific leading-order scaling form (e.g. v_m = 0 forces E_0(n) - Delta_infinity ~ O(n^-2) with no 1/n term, by the quadratic band-minimum structure) -- but only when an explicit dynamical or spatial bridge is documented.

4.2 Type-B [Asymptotic Limit, tag: [->: rate]]

A claim of the form lim_{n->inf} f(n) = 0, where (i) the existence of the limit is established (typically by physical mechanism); (ii) the rate O(n^-p) is determined a posteriori by the specific mechanism; and (iii) the rate is NOT forced by any symmetry or algebraic identity.

Logical force. A Type-B limit constrains only the existence of convergence. The rate must be computed from the mechanism; it cannot be inferred from any Type-A identity without an explicit dynamical or spatial bridge.

4.3 The Mandatory Rules (Anti-Conflation)

Rule #20.1 -- Bridge Requirement

An exact identity (Type-A [=]) cannot be used to dictate the asymptotic rate of a mechanism-determined quantity (Type-B [->: rate]) without an explicit dynamical or spatial bridge. The bridge must identify which Type-A identity constrains which Type-B form coefficient, and the constraint must be derived, not assumed.

Rule #20.2 -- Mandatory Tagging

Every vanishing claim in the VN register, RF-1 through RF-8, must carry an explicit [=] or [->: rate] tag. Untagged vanishing claims are non-compliant and require re-tagging before any framework use.

Rule #20.3 -- Form Constrained, Rate Not

A Type-A identity may constrain the FORM of a Type-B quantity's expansion (e.g. forcing the leading term to be a specific power of 1/n), but only when an explicit bridge is documented. Without the bridge, only the existence of convergence is claimed.

Rule #20.4 -- Default Classification

When in doubt, tag as Type-B [->: rate unknown]. Promotion to Type-A [=] requires explicit derivation or citation of the symmetry / algebraic / representation-theoretic source.

4.4 Retroactive Reclassifications (audit-trail, enforced)

Claim	Old status	New tag	Justification
v_m = 0 at k = pi	(implicit in RF-2)	[=] (Type-A)	Forced by spatial-inversion parity.
E_0(n) - Delta_inf -> 0	(implicit)	[->: 1/n^2] (Type-B; rate bridged from v_m = 0)	Rate forced by Type-A bridge via omega''(pi).
delta(n) -> 0	(mistagged as 1/n in v6.5-RF2; revised to 1/n^2 in v6.5-RF2-REVISED -- both errors)	[->: 1/n^3] (Type-B; no Type-A bridge)	Rate determined by standing-wave boundary density |psi_m(1)|^2 ~ 2 pi^2 (1+xi)^2 / n^3. No Type-A forces the rate.
epsilon_n - Delta_inf -> 0	(implicit)	[->: 1/n^2 leading] (Type-B; rate bridged from v_m = 0)	Bulk-dominated; boundary 1/n^3 is sub-leading.
g_FS = (1/2) Hess(Phi)	(planned in RF-8)	[=] (Type-A)	Kahler-potential identity on projective Hilbert space.
gamma_DL^k -> 0 (detectability contraction)	(implicit in RF-4)	[->: gamma_DL = 1/sqrt(3)] (Type-B; rate from RF-1 sandwich norm)	Mechanism-determined; no Type-A forces the rate.

4.5 Canonical Examples

Example 1 -- Type-A constrains Type-B form via bridge

The exact identity v_m = 0 (Type-A, by parity) constrains the FORM of E_0(n) - Delta_infinity (Type-B) to have leading term ~ alpha/n^2 with NO 1/n term. The bridge is the quadratic band minimum omega(k) = Delta_infinity + (1/2) omega''(pi)(k - pi)^2 + ... and standing-wave quantization k_m = pi m / (n + 2 xi) gives the leading 1/n^2 dependence. The bridge is explicit and derivable.

Example 2 -- Type-B rate NOT constrained by Type-A

The boundary exchange delta(n) = J_eff(n) -> 0 (Type-B) is determined by the standing-wave boundary density |psi_m(1)|^2 ~ 2 pi^2 (1+xi)^2 / n^3 (mechanism-specific). The Type-A identity v_m = 0 does NOT bridge to delta(n)'s rate -- the boundary density depends on the wavefunction structure at x=1, not on the bulk dispersion curvature at k=pi. The rate 1/n^3 must be computed from the boundary MPS tensor.

Example 3 -- Type-A as foundation for Type-B flow

The Hessian identity g_FS = (1/2) Hess(Phi) is itself Type-A (Kahler geometry). It establishes a foundation for the hierarchical RP principle (RF-8), which is a Type-B parabolic gradient flow whose rate is determined by the Hessian spectrum of g_FS.

5. The Hessian Identity Foundation

The Hessian identity g_FS = (1/2) Hess(Phi) is the foundational Type-A tautology on which the framework's metric-flow layer is built. It was established in v6.5-RF8 (Theorem 3.1) and is preserved unchanged in this document. The content is in CHOOSING Phi = 2 log <psi|psi> as the canonical Kahler potential for quantum state space, then DERIVING the framework's structural consequences from this single exact identity.

5.1 The Fubini-Study Metric on Projective Hilbert Space

Let H be a finite-dimensional complex Hilbert space. The projective space CP(H) = (H \ {0}) / C* carries the Fubini-Study metric:

ds^2_FS = (<d psi | d psi> / <psi|psi>) - |<psi|d psi>|^2 / <psi|psi>^2

In holomorphic coordinates [z_0 : ... : z_n] on CP^n, this becomes:

g_{i j-bar} = d_i d_{j-bar} K_FS, where K_FS = log(Sigma |z_k|^2)

5.2 Theorem 3.1 (Hessian Identity, Type-A [=])

Theorem 3.1 (Hessian Identity, Type-A [=]). For the Fubini-Study metric on CP(H): g_FS = (1/2) Hess(Phi), where Phi = 2 K_FS = 2 log <psi|psi>. Status: [Established -- Type-A [=], Class E]. Registered as VN-19.

Proof. In holomorphic coordinates [z_0 : ... : z_n] on CP^n, the Kahler potential is K_FS = log(Sigma |z_k|^2). The Hessian of a function f on a Kahler manifold in complex coordinates is Hess(f)_{i j-bar} = d_i d_{j-bar} f. Setting Phi = 2 K_FS gives:

Hess(Phi)_{i j-bar} = d_i d_{j-bar} (2 K_FS) = 2 d_i d_{j-bar} K_FS = 2 g_{i j-bar}

Hence g_FS = (1/2) Hess(Phi). QED.

Symbolic verification (script: /scripts/verify_hessian_identity.py): CP^1 (Bloch sphere) gives g_{z z-bar} / Hess(Phi)_{z z-bar} = 1/2 (exact rational, confirmed); CP^2 (spin-1 system, 3-component state) gives all components g_{i j-bar} / Hess(Phi)_{i j-bar} = 1/2 (exact rational, confirmed).

Content of the identity. The identity is a tautology of Kahler geometry -- any Kahler metric equals the Hessian of its Kahler potential. The framework's content is in choosing Phi = 2 log <psi|psi> as the canonical potential for quantum state space. From this single exact identity, three structural consequences follow (sections 5.3, 5.4, and 6).

5.3 The Local Algebraic Compatibility Constraint F_MM = 0

The Hessian identity is algebraic and holds identically. However, the physical interpretation -- that g_FS measures the "geometric burden" of state preparation -- requires restricting to the constraint manifold:

M_constraint = { psi in CP(H) : F_MM(psi) = 0 }

where F_MM is the moment-map constraint for the symmetry group G of the physical Hamiltonian H. Concretely:

F_MM(psi) = (H psi - E_0 psi) / ||psi||^2 (modulo ground-state projection)

F_MM = 0 iff psi is a ground state of H (in the appropriate representation). More generally, F_MM = 0 specifies the algebraic compatibility of psi with the symmetry structure. The moment-map equation is a symmetry-forced algebraic identity; registered as VN-20, Type-A [=].

5.4 The Geometric Burden B_geom = Var(g_FS)

Definition 3.2 (Geometric Burden, Type-A [=]). The geometric burden of a state psi in M_constraint is:

B_geom(psi) = Var(g_FS)|_psi = int_{CP(H)} |g_FS(psi) - <g_FS>|^2 d mu_CP(psi)

where the variance is computed over the projective Hilbert space with respect to the FS-induced measure d mu_CP.

Physical interpretation. B_geom measures how non-uniform the FS metric is around psi. A state with B_geom = 0 is "geometrically neutral" -- the metric is uniform in its neighborhood. States with B_geom > 0 carry "geometric stress" that must be relaxed by the parabolic flow of section 6.

Bounded-below property. B_geom >= 0 trivially (variance is non-negative), with equality iff g_FS is locally constant. This is the variational structure required by Protocol #16.

SECTION 5 VERDICT: The Hessian identity is [Established -- Type-A [=], Class E, VN-19]. It is preserved unchanged from v6.5-RF8. The constraint-manifold construction F_MM = 0 is [Established -- Type-A [=], VN-20]. The geometric burden B_geom is bounded below by 0 (variance), satisfying Protocol #16.

6. The Hierarchical RP Principle

The Hierarchical RP (Robustness Principle) is the natural gradient flow on (M_constraint, g_FS) generated by the Hessian identity foundation. It is preserved unchanged from v6.5-RF8 (Theorem 4.1) and provides the framework's mechanism for relaxing geometric stress. The principle decomposes into a Type-A instantaneous layer (constraint satisfaction) and a Type-B mechanism-determined layer (metric relaxation), making robustness a sequential layer optimization rather than a simultaneous relaxation.

6.1 The Parabolic Gradient Flow (Type-B [->: rate])

The natural gradient flow on (M_constraint, g_FS) is:

d_t psi = - grad_{g_FS} B_geom(psi) + lambda * grad_{g_FS} (F_MM(psi))

where lambda is a Lagrange multiplier enforcing F_MM = 0 along the flow. This is a Type-B mechanism-determined flow; its rate is determined by the Hessian spectrum of g_FS on M_constraint.

6.2 Layer-Sequential Decomposition (Theorem 4.1)

Theorem 4.1 (Parabolic Flow Structure, Type-A + Type-B). The flow decomposes into two sequential layers: (1) Layer 1 (Type-A, instantaneous): d_t F_MM(psi) = - L_F * F_MM(psi) -> 0, rate determined by the symmetry-algebra spectrum of L_F. (2) Layer 2 (Type-B, mechanism-determined): d_t B_geom(psi) = - <grad B_geom, grad B_geom>_{g_FS} -> 0, rate O(t^{-1/2}) or O(e^{-lambda_min t}) depending on the Hessian spectrum of g_FS. Status: [Established].

Proof sketch. The constraint F_MM = 0 defines a smooth submanifold M_constraint subset CP(H) by the implicit function theorem (using non-degeneracy of the moment map, a Type-A [=] property). The restriction of g_FS to M_constraint is again Kahler (closedness of g_FS is preserved). On this Kahler submanifold, B_geom is a bounded-below functional whose gradient generates a well-posed parabolic flow. The flow's leading behavior is determined by the spectrum of Hess(B_geom) on M_constraint, which is Type-B (mechanism-determined). QED.

6.3 Layer-Sequential Optimization Algorithm

The flow has the explicit layer-sequential structure:

Step 1: Project psi onto M_constraint (F_MM = 0). [Well-defined by Type-A non-degeneracy of the moment map.]
Step 2: On M_constraint, evolve psi by the B_geom gradient flow. [Well-defined because B_geom is bounded below (by 0), Type-A.]
Step 3: The limit psi_infty satisfies F_MM = 0 AND grad B_geom = 0. [Variational characterization of the ground state.]

6.4 Connection to RF-4 (Detectability Lemma)

The detectability lemma (RF-4 / C-02) is the Type-B dynamical manifestation of the Type-A kinematic layerability:

Type-A kinematic layerability (Theorem 5.1). By Vizing's theorem, the edge-coloring number of a graph G with bounded causal degree Delta satisfies chi'(G) <= Delta + 1. Therefore, the interaction Hamiltonian H =

Sigma_e h_e admits a decomposition into at most g = Delta + 1 commuting layers: H = H_1 + H_2 + ... + H_g, [H_i, H_j] = 0 for i != j. This is Type-A [=], follows from Vizing's theorem (combinatorial identity), independent of physical mechanism.

Type-B reconciliation contraction. On M_constraint with the parabolic flow, the error residual after k steps of layer-sequential projection satisfies ||error_k||^2 ~ gamma_DL^{2k} -> 0, where gamma_DL = 1/sqrt(3) is the DL contraction constant (VN-17, Type-B [->: rate]). The detectability lemma is the Type-B dynamical manifestation: the parabolic flow contracts errors at the rate gamma_DL^k -> 0, ensuring any initial state with F_MM ~ 0 converges to the ground-state manifold exponentially fast.

Status of C-02. The 1D specialization is the Aharonov-Arad-Landau-Vazirani lemma (NJP 13, 113043, 2011, arXiv:1011.3445). The general-graph version follows from the Type-A layerability plus the Type-B contraction. The bibliographic lock on C-02 is preserved (Protocol #19).

6.5 The Reconciliation Cascade

[RF-8: HESSIAN IDENTITY] g_FS = (1/2) Hess(Phi) [Type-A Identity, VN-19]
[TRIANGULAR LAYER ORDERING] Local Algebra -> Spatial Metric -> Gravity
[RF-4: CAUSAL LAYERABILITY] Bounded Causal Degree => g Commuting Matchings
[DETECTABILITY CONTRACTION] Error Residual ~ gamma_DL^{2k} -> 0 [Type-B Flow]

7. The AKLT Arc: Six-Round Substrate Investigation

This section documents the six-round substrate investigation that constitutes the AKLT arc. The arc's goal was to attempt a concrete realization of the projection picture (the framework's hypothesis that relational information survives strain via a bulk-to-boundary projection mechanism) on the controlled substrate of the AKLT spin-1 chain. The arc produced a negative verdict on every dynamical form the substrate could host; the verdict is permanent scientific capital and is recorded here with the decisive data tables and trajectories that drove each round's conclusion.

Each round is presented in the same four-part structure: (i) Setup, (ii) Decisive data (the 2-3 tables that actually drove the verdict), (iii) Verdict, (iv) Carry-forward to next round. The reader can audit each decision without opening external files; full data is in the JSON result files in /home/z/my-project/download/.

7.0 Substrate & Burden Dial

Substrate: AKLT spin-1 chain, H_AKLT = Sigma_{j=0..N-2} [1/2 S_j.S_{j+1} + 1/6 (S_j.S_{j+1})^2], with periodic boundary conditions (PBC) for delta/epsilon/zeta/eta/eta-extreme and open boundary conditions (OBC) for Option A. Default system size N=10; N=12 used in two-size tests.

Burden dial. The arc uses two distinct strain burdens, distinguished by which physical channel they couple to:

V_strain = + h * sum_j (S^z_j)^2 [existence-channel burden; drives Haldane -> large-D transition]

V_FM = - J * sum_j S_j.S_{j+1} [temporal-channel burden; drives Haldane -> Ising-FM transition]

Arc rounds. The six rounds are: **delta** (V_strain dial, primary observables); **epsilon** (joint distance x direction composite); **zeta** (V_FM temporal burden); **eta** (two-size floor test, 1/N scaling); **eta-extreme** (h >> h_c, ground-state floor); **Option A** (OBC edge-state test, the only unsearched location). All rounds use multiplet-corrected gaps (singlet-channel gap when the ground state is in a nontrivial SU(2) sector); see section 10 for the multiplet-blindness artifact this corrects.

7.1 delta -- Strained String Burden Dial

Setup. AKLT N=10 PBC, V_strain = +h * sum_j (S^z_j)^2, h in {0, 0.1, ..., 2.0} (21 points). Per h: gap (multiplet-corrected), existence <(S^z)^2>, distance <S^z_0 S^z_3>, direction <S^z_0 S^z_1>, string <O_string(0,3)>. Four predictions tested: (P1) existence smooth at the gap minimum; (P2) L_0 floor realized; (P3) position+scale survive; (P4) string order survives.

Decisive table: burden dial trajectory. Selected h values showing the trade-off between gap (R4) and existence (R3):

h	gap	existence	direction (ell=1)	string (ell=3)	verdict
0.0	0.350	0.667	-0.444	-0.444	baseline (AKLT vacuum)
0.5	0.137	0.475	-0.213	-0.206	partial: gap closing, structure decaying
0.9	0.073	0.325	-0.108	-0.071	gap minimum (h_c); structure 50% baseline
1.0	0.074	0.284	-0.092	-0.050	post-min: gap opens, structure decays faster
1.3	0.168	0.153	-0.056	-0.011	gap > 2x min; structure < 25% baseline
2.0	0.752	0.044	-0.020	-0.0003	large-D phase; structure -> 0

Source: wp5c_delta_strained_string_results.json (full 21-point scan).

Decisive finding (SPT-fragility diagnosis). The four-prediction falsification:

P1 (existence smooth at gap min): PASS. Existence = 0.325 at h=0.9 (gap min), 0.284 at h=1.0. Smooth.
P2 (L_0 floor): FAILS POST-ARC. Gap minimum 0.073 at h=0.9 is a 1/N conformal artifact (eta two-size test). Not a substrate floor.
P3 (position+scale survive): PARTIAL FAIL. Sum ratio 0.383 at h_c; structure decays to ~25% baseline by h=1.3.
P4 (string survives): FAIL. String -0.444 -> -0.011 by h=1.3 (97% loss); no protection.
VERDICT: SPT-invariant roles are NOT burden-robust on this substrate.

Carry-forward to epsilon. The user's correction: distance and direction are complementary (jointly define position); pairing should give robustness. The next round tests the joint distance x direction composite as a candidate rescue for the position channel.

7.2 epsilon -- Joint Distance x Direction Composite

Setup. AKLT N=10 PBC, V_strain (same as delta), per-h measurement of: existence, distance <S^z_0 S^z_ell> (ell=1,2,3,5), direction <S^z_0 S^z_1>, joint_product = distance * direction (the position composite), sum_position = distance + direction. Test whether anti-correlation between distance and direction channels gives robustness by cancellation.

Decisive table: complementarity at h_c. The joint composite cancellation relief mechanism (ell=2 is where channels anti-correlate):

h	ell	distance	direction	joint = d * D	sum = d + D
0.0	3	-0.050	-0.444	+0.050	-0.494
0.5	3	+0.004	-0.206	-0.008	-0.202
0.9	3	-0.013	-0.071	+0.001	-0.084
0.5	2	-0.028	-0.211	+0.027	-0.239
0.9	2	-0.030	-0.083	+0.034	-0.113
1.3	3	-0.005	-0.011	+0.0001	-0.016

Source: wp5c_epsilon_joint_composite_results.json.

Decisive finding (ell=2 complementarity). At ell=2, the joint product *doubles* at h_c relative to baseline (0.034 vs 0.018 at h=0); at ell=3, the joint product is also enhanced (0.001 vs -0.005 at h=0). The complementarity is real: when distance and direction anti-correlate, their product cancels the leading-order decay. This is robustness-by-anti-correlation, not robustness-by-protection. It is bound to the regime where channels anti-correlate (ell >= 2); at ell=1 (nearest-neighbor) the channels are degenerate and the mechanism does not apply.

EPSILON VERDICT: Complementarity mechanism ACCEPTED. Joint distance x direction composite shows ~2x enhancement at h_c for ell=2, confirming robustness-by-anti-correlation. Scope sharpened: 'robustness by anti-correlation' (not protection); bound to regime where channels anti-correlate (ell >= 2). ell >= 2 channel-distinctness fact REGISTERED for F5 channel-structure derivation; ell=1 degeneracy is load-bearing.

Carry-forward to zeta. Epsilon shows the existence channel (V_strain) has complementarity at ell >= 2 but no protection at ell=1. The next round tests a different burden (V_FM, temporal channel) to see whether R3+R4 can be simultaneously realized when the burden couples to a different physical channel.

7.3 zeta -- Temporal Burden (FM Heisenberg)

Setup. AKLT N=10 PBC, V_FM = -J * sum_j S_j.S_{j+1}, J in {0.0, 0.1, ..., 2.0} (21 points). Drives the Haldane -> Ising-FM transition. Same four observables as delta (existence, distance, direction, sum_position). Three predictions tested: (P1) existence HIDDEN not COLLAPSED under temporal burden; (P2) gap min = L_0 floor; (P3) position+scale FULL survival.

Decisive table: V_FM gap and survival trajectory.

J	gap (naive)	existence	sum_pos (ell=3)	verdict
0.0	0.350	0.667	-0.494	baseline (Haldane)
0.3	~0 (manifold)	0.671	-0.035	Haldane -> FM transition; gap artifact (see eta)
0.5	~0 (manifold)	0.690	+0.117	sign flip in sum_pos; FM order emerging
0.7	~0 (manifold)	0.679	+0.545	FM phase; sum_pos recovers > baseline
1.0	~0 (manifold)	0.818	+0.989	deep FM; full recovery
1.5	~0 (manifold)	0.661	+0.463	deep FM; recovery persists
2.0	~0 (manifold)	0.683	+0.579	deep FM; recovery persists

Source: wp5c_zeta_temporal_burden_results.json.

Decisive finding (R3 recovery, R4 manifold artifact). P1 PASS: existence stays at ~0.67-0.91 (hidden, not collapsed) across the entire J scan. P3 PASS: sum_position recovers to 0.99 at J=1.0 (above baseline) with a sign flip (AFM -> FM order emerging in the temporal sector). P2 FAILS: gap = 0 exactly throughout the FM phase is the Lanczos within-manifold splitting (see eta Part 1 for the multiplet diagnostic that resolves this).

ZETA VERDICT: R1 (existence HIDDEN burden-independent): PASS. R3 (position+scale survive under temporal burden): FULL PASS with recovery (sum_pos 0.99 at J=1.0, above baseline). R4 (L_0 floor): STRUCK. Gap=0 is the Lanczos within-manifold artifact (FM SU(2) multiplet); true finite-N gap to next multiplet is small but nonzero (eta diagnostic: ~0.001 at J=0.3, ~0.08 in deep FM), vanishing only as N -> inf. R4 cannot be drawn from the artifact.

Carry-forward to eta. Zeta's R4 strike is mechanism-level (Lanczos manifold blindness, not a sign error). But this leaves the framework with a tension: the burden that gives full R3 recovery (V_FM) violates R4, and the burden that may satisfy R4 (V_strain, delta's gap minimum 0.073) gives only partial R3. The next round runs the two-size test under V_strain to settle whether R4's floor is substrate physics or finite-N artifact.

7.4 eta -- Two-Size Floor Test (V_strain 1/N scaling)

Setup. Two parts. Part 1: degenerate-manifold diagnostic at J=0.3 (N=10) under V_FM to resolve zeta's struck gap=0. For each (N, J) under V_FM: get k=12 low eigenvalues, compute <psi_i | S_total^2 | psi_i> for each, identify SU(2) sector (S(S+1) ~ 0 singlet, 2 triplet, 6 quintet, ...). gap_corrected = E(lowest non-ground-multiplet state) - E(ground). Part 2: V_strain two-size scan (N=10, N=12) at h in {0, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95, 1.0, 1.1, 1.3, 1.5}. For each (N, h): gap_corrected (singlet channel), existence, distance, direction, sum_position. Find L_0 floor (gap_min) per N. Test 1/n^2 scaling: gap_min(N=10) - gap_min(N=12) vs beta * (1/100 - 1/144). Verdict: artifact or substrate physics.

Part 1: V_FM multiplet diagnostic (resolves zeta)

Decisive table: SU(2) sector identification at selected J values (N=10, k=12, tol=1e-10).

J	S_g (ground)	gap_naive	gap_corrected	to S =	classification
0.00	0 (singlet)	0.3501	0.3501	1	consistent
0.10	0 (singlet)	0.3119	0.3119	1	consistent
0.30	1 (TRIPLET)	0.0000	0.001231	0	MANIFOLD ARTIFACT
0.70	10 (FM, S=N)	0.0000	0.0764	9	MANIFOLD ARTIFACT
1.00	10	0.0000	0.1910	9	MANIFOLD ARTIFACT
1.50	10	0.0000	0.3820	9	MANIFOLD ARTIFACT
2.00	10	0.0000	0.5729	9	MANIFOLD ARTIFACT

Source: wp5c_eta_two_size_floor_results.json (Part 1).

Decisive finding (Part 1). Zeta's gap=0 at J=0.3 is a Lanczos manifold artifact: the ground state at J=0.3 is in the S=1 triplet sector (3-fold degenerate at SU(2) symmetric point). Lanczos returns the within-triplet splitting, which is exactly zero by SU(2) symmetry. The TRUE finite-N gap to the lowest DIFFERENT sector (S=0 singlet) is 0.001231 -- small but nonzero. This is the finite-N regularization of the Haldane -> FM critical point (J_c ~ 0.3 on N=10), where the gap vanishes in the thermodynamic limit but is regularized to O(v*pi/N) on finite chains. At J >= 0.7 (deep FM phase, S_g=10 = fully polarized S=N multiplet): gap_naive=0 is the within-FM-multiplet splitting (21-fold degenerate at SU(2) symmetric point for S=N=10). The TRUE Goldstone gap (to S=N-1 multiplet) is finite: 0.076, 0.191, 0.382, 0.573 at J=0.7, 1.0, 1.5, 2.0 respectively.

Part 2: V_strain L_0 floor two-size test

Decisive table: gap_min scaling N=10 vs N=12 (PBC, V_strain).

Observable	N=10	N=12	ratio N12/N10	verdict
Baseline gap (h=0)	0.3501	0.3501	1.0001	STABLE (AKLT exact solvability)
Gap at h=0.9 (claimed h_c)	0.0732	0.0619	0.8454	~1/n (conformal critical)
L_0 floor (gap_min over scan)	0.0726 at h=0.95	0.0608 at h=0.95	0.8379	1/n SCALING (conformal critical)
Predicted under 1/n^2	0.0726	0.0504	0.6944	Distance 0.14 -- too far
Predicted under 1/n	0.0726	0.0605	0.8333	Distance 0.005 -- EXACT match
Predicted under stable	0.0726	0.0726	1.0	Distance 0.16 -- too far

Source: wp5c_eta_two_size_floor_results.json (Part 2).

Decisive finding (Part 2). The V_strain "Haldane -> large-D" transition at h_c ~ 0.9-0.95 is a CONFORMAL critical point (Lorentz-invariant CFT). The gap at the critical point scales as v*pi/N (conformal gap formula), giving the observed 1/N scaling. R4's L_0 floor (0.0732 in delta) is the finite-N regularization of this critical point, NOT substrate physics. R4 is STRUCK for V_strain (consistent with zeta's V_FM strike). The auditor's hypothesis is fully vindicated: "If L_0 is finite-N regularization, R4 fails in the thermodynamic limit for every burden -- the floor is not a substrate feature but an artifact class." The black-hole picture's "L_0 floor" was the finite chain all along.

ETA VERDICT (Parts 1 + 2): Part 1: Zeta's V_FM gap=0 at J=0.3 CONFIRMED as Lanczos manifold artifact (within-triplet splitting by SU(2) symmetry). True finite-N gap 0.001231 (singlet channel), vanishing as N -> inf. Part 2: V_strain L_0 floor (delta's 0.073) is a 1/N conformal artifact (distance to 1/n: 0.005; to

1/n^2: 0.14). R4 STRUCK for BOTH burdens. R4 is not a substrate feature; it is finite-N regularization. The black-hole picture's L_0 floor was the finite chain all along.

Carry-forward to eta-extreme. With R4 fully struck at the critical point, the user proposed distinguishing the critical-point L_0 floor (struck, 1/N artifact) from a "ground-state L_0 floor under extreme strain" that might survive as a substrate feature in extreme regimes (BH-like). The next round tests this by scanning h >> h_c.

7.5 eta-extreme -- Ground-State Floor Under Extreme Strain

Setup. V_strain = +h * sum(S^z)^2, AKLT N=10 and N=12, PBC. h_extreme = {1.5, 2.0, 3.0, 5.0, 10.0} (well beyond h_c ~ 0.95). Per (N, h): k=10 lowest eigenvalues, gap, existence, distance, direction, joint, sum_position. Three tests: (1) substrate floor stability (gap ratio N12/N10); (2) gap growth with h (expected gap -> h as h -> inf); (3) position+scale survival in extreme-strain ground state.

Decisive table: extreme-strain substrate floor stability.

h	gap N=10	gap N=12	ratio	verdict
1.5	0.309	0.305	0.9880	STABLE
2.0	0.752	0.752	0.9996	STABLE
3.0	1.716	1.716	1.0000	STABLE
5.0	3.693	3.693	1.0000	STABLE
10.0	8.679	8.679	1.0000	STABLE

Decisive table: position+scale survival in extreme-strain ground state (N=10).

h	existence	distance	direction	sum_pos	verdict
1.5	0.098	-0.002	-0.003	-0.005	R3 partial (gap > 0 but structure decaying)
2.0	0.044	~0	~0	-0.0006	R3 collapsing
3.0	0.016	~0	~0	~0	R3 collapsed
5.0	0.005	~0	~0	~0	R3 fully collapsed
10.0	0.001	~0	~0	~0	R3 endpoint: all-m=0 product state

Source: wp5c_eta_extreme_strain_results.json.

Decisive finding (user's distinction confirmed; R3/R4 tension). User's category distinction is REAL: extreme-strain floor IS a substrate feature (stable with N across h >> h_c), distinct from the critical-point floor (1/N artifact, struck in eta). BUT the floor does NOT "prevent singularity" in the original R4 sense. The singularity (gap=0) is at the critical point h_c; in the extreme-strain regime (h > h_c), the system is in a DIFFERENT GAPPED PHASE (large-D phase) with finite gap by construction. The floor isn't preventing the singularity; it's the regular gap of a different phase. CRITICAL TENSION with R3: in the extreme-strain limit (h -> inf), the ground state becomes the all-m=0 product state, where ALL relational observables (distance, direction, joint, sum) -> 0. So R4-restored (extreme-strain floor) comes at the cost of R3 collapse (position+scale fully collapse in the same limit). This is the SAME R3/R4 tension the auditor identified: on this substrate, you can have either (a) near critical: R4 finite-N artifact (struck), R3 partial survival (sum ratio 0.383 at h_c); or (b) extreme strain: R4 substrate floor (real, stable), R3 collapse to 0 (position+scale -> 0 in all-m=0 ground state). But you cannot have both R3-survival AND R4-floor simultaneously.

ETA-EXTREME VERDICT: User's distinction ACCEPTED: extreme-strain floor IS a substrate feature (stable with N, gap -> h as h -> inf, distinct from the critical-point 1/N artifact). R4 (L_0 floor): STRUCK at critical point (1/N artifact, eta Part 2); RESTORED in extreme-strain regime (substrate feature, stable with N). R3 (position+scale): PARTIAL near critical (0.383 at h_c); FULL COLLAPSE in extreme strain (-> 0 in all-m=0 product state). R3/R4 tension: CONFIRMED persistent across both regimes; cannot be simultaneously realized on AKLT substrate with V_strain. The trade-off is monotone (no interior extremum hiding both).

Carry-forward to Option A. Eta-extreme completes the characterization of the V_strain dial: gap vs. information is monotone, no interior extremum, h -> inf endpoint has all channels cloaking together at 10^-3. The only unsearched location is the BOUNDARY (OBC), where AKLT has natural S=1/2 edge modes -- the SPT-edge form of the user's hypothesized 'singularity boundary projection' mechanism. The next round tests whether these edge modes carry the position+scale composite that bulk PBC could only partially sustain.

7.6 Option A -- OBC Edge-State Test (the only unsearched location)

Setup. H(h) = H_AKLT(OBC) + h * V_strain. N in {10, 12}, k_eig=6 (4 for multiplet diagnostic + 1 for bulk_gap). h in {0, 0.1, 0.3, 0.5, 0.7, 0.9, 0.95, 1.0, 1.1, 1.3, 1.5, 2.0, 3.0, 5.0, 10} (15 points). Primary observables: edge_splitting = E[3] - E[0] (4-fold ground-state manifold spread); bulk_gap = E[4] - E[0] (4-fold top -> first bulk excitation). Edge observables (on ground state): edge_existence = <(S^z_0)^2 + (S^z_{N-1})^2>; edge_corr = <S^z_0 S^z_{N-1}>; edge_string = <O_string(0, N-1)>. Bulk observables (direct comparison, sites 0 and 3): bulk_existence = <(S^z_3)^2>; bulk_corr = <S^z_0 S^z_3>; bulk_string = <O_string(0, 3)>. Multiplet diagnostic forced from first data point.

Part 2: Edge splitting scaling class (primary observable)

Decisive table: edge splitting N=10 vs N=12 across the full h scan.

h	es_N10	es_N12	ratio	scaling class
0.00	0.370	0.364	0.985	STABLE (bulk Haldane gap, h=0)
0.10	3.5e-04	9.2e-05	0.265	STRONG finite-N (multiplet splitting)
0.30	7.4e-03	3.5e-03	0.473	STRONG finite-N
0.50	3.3e-02	2.2e-02	0.649	~1/n^2
0.70	8.2e-02	6.2e-02	0.759	~1/n
0.90	1.65e-01	1.33e-01	0.808	~1/n (conformal critical)
0.95	1.95e-01	1.60e-01	0.818	~1/n (conformal critical)
1.30	4.17e-01	3.59e-01	0.859	~1/n
2.00	9.71e-01	9.14e-01	0.941	STABLE (large-D)
10.00	8.86e+00	8.96e+00	1.011	STABLE (large-D)

Source: wp5c_optionA_edge_test_results.json.

Decisive finding (Part 2). Edge splitting scaling class COMPLETELY FOLLOWS BULK GAP SCALING CLASS. Small h (h<0.5): exponential finite-N (multiplet splitting, xi ~ 1.6). Critical h (h ~ 0.5-1.5): 1/n conformal (same as bulk critical point). Extreme h (h >= 2.0): stable (same as bulk large-D phase floor). The edge has NO scaling class of its own -- the edge's "own gap" evolves in lockstep with the bulk gap.

Part 4: Edge vs bulk observable persistence (projection test core)

Decisive table: edge vs bulk existence (N=10, key h values). The projection test: does edge survive when bulk composites are at 1e-3?

h	edge_exist	bulk_exist	e/b ratio	interpretation
0.00	1.333	0.667	2.000	SPT-edge protection PRESENT (clean AKLT)
0.10	1.176	0.627	1.876	Protection still strong
0.30	0.887	0.548	1.620	Protection decaying
0.50	0.645	0.469	1.374	Protection further decay
0.70	0.450	0.395	1.140	Protection near loss
0.90	0.295	0.315	0.936	Protection gone
1.00	0.232	0.268	0.867	INVERTED: bulk > edge
1.30	0.115	0.141	0.815	Inverted; both decaying
2.00	0.038	0.043	0.882	Both at ~4e-2
5.00	0.005	0.005	0.962	Both at ~5e-3
10.00	0.0012	0.0012	0.982	Both at ~1e-3; fully equivalent

Source: wp5c_optionA_edge_test_results.json (Part 4).

Decisive finding (Part 4). h=0: edge observables = 2 x bulk (SPT-edge protection EXISTS but only in clean AKLT). h growing: edge/bulk ratio -> 1 (protection lost, h ~ 1 inverts to bulk > edge). h -> inf: edge = bulk (fully equivalent, no protection). This is the delta fragility result REPLICATED ON THE BOUNDARY. Projection test (edge > 0.05 while bulk < 0.05): FALSE at h=1.3 (both at ~0.13). At h=10 endpoint: edge/bulk = 0.98 (~1.0), all relational channels co-cloak at 10^-3.

OPTION A VERDICT: SPT-edge form of the singularity boundary projection IS NOT REALIZED on this substrate. (i) Edge splitting scaling = bulk gap scaling (no independent boundary floor). (ii) Edge observables = bulk observables evolving with h (protection only at h=0 clean AKLT; h ~ 1 loses; h -> inf fully equivalent). (iii) h -> inf endpoint: edge/bulk ~ 1.0, all relational channels co-cloak at 10^-3. The delta fragility result replicates on the boundary -- SPT-edge protection does not survive V_strain. The trade-off stands as a property of this substrate. The AKLT laboratory has answered its final question in the negative for every location it possesses.

Carry-forward. Refinement: the projection structure EXISTS at h=0 (SPT-edge protection: edge = 2 x bulk), but it is FRAGILE (h ~ 1 destroys it, not robust against V_strain), consistent with the AdS/CFT-class analogy reminder: SPT-edge is one realization form's direct test; the negative result rules out THIS form of the projection but does not rule out other forms (topological encoding, quasiparticle, etc.) not tested here. But SPT-edge is the only natural boundary-projection candidate the AKLT substrate possesses; without it, the substrate as a whole does not host boundary projection. The conversation moves to substrate change (Appendix A).

7.7 Synthesis: The Arc as a Single Picture

The six rounds of the AKLT arc form a single coherent investigation, not six independent tests. Reading them as a single picture makes the through-line visible: the substrate answered the framework's question at every location and regime it could be asked, and the answer was uniformly negative. The negative answer is not a failure of the framework's question; it is the substrate's response to it.

The logical structure of the arc

The arc has a deductive structure: each round either falsifies a hypothesis the previous round left open or characterizes an artifact the previous round's verdict depended on. **Delta** established the burden dial's basic trajectory and falsified the burden-robustness of SPT-invariant roles. **Epsilon** attempted a rescue via the joint composite and found a real but bounded complementarity mechanism (ell >= 2 anti-correlation), sharpening the failure to the ell=1 degeneracy. **Zeta** switched to the temporal channel and found R3 recovery but R4 struck via gap=0, which the next round would resolve as an artifact. **Eta** ran the two-size test that settled R4 as a 1/N conformal artifact under V_strain and resolved zeta's gap=0 as a Lanczos manifold artifact under V_FM. **Eta-extreme** confirmed the user's distinction (extreme-strain floor is real, distinct from the critical-point floor) and characterized the R3/R4 tension as persistent across both regimes. **Option A** tested the only unsearched location (the boundary) and found the delta fragility replicated there: SPT-edge protection is fragile, edge observables co-fold with bulk observables under V_strain, no role-transfer occurs.

The substrate's complete trade-off surface

The arc produced a complete map of the substrate's trade-off surface: the relationship between gap (R4) and relational structure (R3) across the full burden dial. The map has three regions, each characterized with machine precision. **Critical region (h ~ h_c ~ 0.9-0.95):** gap is small (0.07 at minimum), relational structure is partial (sum_pos ratio 0.383 at h_c, existence 0.325), and the gap minimum is a 1/N conformal artifact (struck by eta two-size test). **Extreme-strain region (h >> h_c, h > 2):** gap is large (~0.31 to ~8.7, growing linearly with h), relational structure is collapsed (sum_pos -> 0, existence -> 0), and the gap is a stable substrate feature (eta-extreme confirmed). **Small-strain region (h < 0.5):** gap is near baseline (0.35 to 0.14), relational structure is near baseline (existence 0.67 to 0.48), and the SPT-edge protection (Option A measured edge = 2 x bulk) is intact at h=0 but decaying with h.

The trade-off is **monotone**: gap grows while structure decays, with no interior extremum where both are simultaneously realized. This is the arc's most important structural finding: the substrate's R3/R4 tension is not a calibration issue (the instruments are correctly measuring what is there) but a substrate property (the substrate does not admit a regime where both R3 and R4 are simultaneously realized). The monotonicity rules out the picture's "interior sweet spot" hypothesis (that some intermediate h would preserve both) and forces the framework to look elsewhere.

The measured endpoint (h -> infinity)

The arc measured the h -> infinity endpoint directly. At h=10 (N=10 and N=12, PBC and OBC), the substrate is in the all-m=0 product state, and all relational observables (existence, distance, direction, joint, sum, edge_existence, edge_corr, edge_string, bulk_string) co-cloak at 10^-3 or below. The edge/bulk ratio is ~0.98 (essentially 1.0); there is no boundary-bulk distinction at the endpoint. The endpoint is the cleanest test of the picture's projection mechanism: if the mechanism existed, the endpoint would be where it should most cleanly separate (the substrate is at its simplest, the all-m=0 product state). The data shows no separation; the endpoint confirms the picture's absence on this substrate, at the cleanest point the substrate can offer.

Why the substrate answered "no" everywhere

The primitive-vs-emergent axiom (section 3) gives the structural explanation. The AKLT substrate's primitive-observable algebra is built from spin operators (S^z, S^+, S^-) and their correlations, all defined site-locally without reference to a coarse-graining equivalence class. The arc treated these as primitives because that is what the substrate hands you. But RCF's foundational commitment is that physical observables are always emergent from the relational coarse-graining. The substrate's primitives therefore cannot serve as the framework's emergents directly; the substrate's primitives and the framework's emergents live in different algebras, and the projection picture's bulk-to-boundary isomorphism cannot be constructed across this gap. The negative verdict at every location is the substrate's consistent refusal to host a projection it cannot, structurally, host.

This is why the next substrate must be selected by the primitive-vs-emergent criterion (section 3.4 / Appendix A): the criterion operationalizes the structural correction the arc surfaced. A substrate whose primitive-observable algebra admits an emergent interpretation within RCF -- i.e. whose primitives are (close to) emergents of a relational coarse-graining -- would not have the structural gap that produced the AKLT arc's negative verdict. The 2D SPT substrate (Appendix A, candidate (d)) is the cleanest such candidate: its topological boundary modes are kinematically protected (phase-independent), satisfying all three conditions of the substrate-evaluation criterion.

SECTION 7 SYNTHESIS: The six rounds form a single deductive investigation with a uniformly negative verdict. The substrate's trade-off surface is fully mapped (monotone, no interior extremum). The h -> infinity endpoint confirms the picture's absence at the cleanest point the substrate can offer. The primitive-vs-emergent axiom (section 3) gives the structural explanation: the substrate's primitives and the framework's emergents live in different algebras. The next substrate must be selected by the primitive-vs-emergent criterion.

8. The Picture's Boundary Statement (Scoping Theorem)

The projection picture is the framework's hypothesis that relational information survives strain via a bulk-to-boundary projection mechanism -- an isomorphism between the substrate's bulk algebra and a boundary algebra that decouples when the bulk is strained. The AKLT arc was an attempt to realize this picture concretely on a controlled substrate. The arc's verdict is negative on every dynamical form the AKLT substrate could host; the picture therefore needs scoping. This section states the scoping theorem formally, identifying what survives, what is falsified, and what the next substrate must supply.

8.1 What Survives

The picture's survival is restricted to its kinematic form -- the standard-GR cloaking asymmetry, in which bulk observables are operationally inaccessible from outside a horizon while boundary observables remain accessible. This is the form of the picture that does not depend on a dynamical bulk-to-boundary projection mechanism; it is built into the substrate's geometry, not into a phase the substrate happens to occupy. The kinematic form is the minimum content of the projection picture that the framework continues to endorse.

Specifically, the surviving content is: (i) the asymmetric accessibility of bulk vs. boundary observables, characteristic of standard GR horizons; (ii) the asymmetry is kinematic (built into the substrate's definition) rather than dynamical (produced by a phase or by an SPT protection); (iii) the asymmetry does not require a separate boundary algebra distinct from the bulk algebra -- the boundary observables are simply the bulk observables that survive the horizon constraint.

8.2 What Is Falsified

Every dynamical form the AKLT substrate could host is falsified. Specifically: (i) static metric emergence is blocked by VN-66 (the metric is not emergent from operator correlations in the AKLT vacuum); (ii) simple rate-based emergence is blocked by exponential AKLT correlations (the metric is not a function of any single rate of correlation decay); (iii) bulk relational protection fails under strain (delta shows structure decays to ~25% baseline by h=1.3; epsilon shows anti-correlation robustness only at ell >= 2, not protection); (iv) boundary SPT-edge protection fails under V_strain (Option A shows edge = 2 x bulk at h=0 but edge = bulk at h > 1, all channels co-cloak at h -> inf).

8.3 Boundary Statement (Scoping Theorem)

SCOPING THEOREM (Picture's Boundary Statement). The projection picture survives in the framework ONLY as the standard-GR cloaking asymmetry (kinematic, phase-independent). It is FALSIFIED in every dynamical form the AKLT substrate could host. The picture's survival on future substrates requires a substrate whose boundary protection is kinematic rather than SPT-dynamical. The substrate-change fork (Appendix A) operationalizes this requirement as the substrate-evaluation criterion (section 3.4).

8.4 Substrate-Evaluation Criterion (operationalized)

The scoping theorem's requirement -- a substrate whose boundary protection is kinematic rather than SPT-dynamical -- is operationalized by the substrate-evaluation criterion stated in section 3.4 and applied to the four candidate substrates in Appendix A. The criterion has three conditions: (i) every primitive observable in A_prim(S) must be expressible as a function of C_L-class observables (the primitive-emergent admissibility test); (ii) C_L must admit a nontrivial boundary-bulk decomposition (boundary algebra not identical to bulk algebra); (iii) the decomposition must be kinematic (phase-independent) rather than SPT-dynamical (phase-dependent). The AKLT substrate satisfies (i) but fails (ii) and (iii); the next substrate must satisfy all three.

8.5 What the Next Substrate Must Supply

The next substrate must supply three things the AKLT substrate could not. **First**, a kinematic boundary-bulk decomposition: a boundary algebra distinct from the bulk algebra, defined by the substrate's structure rather than by a phase it occupies. **Second**, robustness of the boundary observables under the substrate's strain burden (whatever form that takes): the boundary observables must not co-fold with the bulk observables when strain is applied. **Third**, the substrate's primitive-observable algebra must admit an emergent interpretation within RCF -- the primitive-emergent gap must be small (ideally zero) so that the substrate's primitives are (close to) emergents of the relational coarse-graining. The four candidate substrates in Appendix A are evaluated against these three requirements.

9. The Four Permanent Facts (Standing Scientific Capital)

The AKLT arc leaves behind four permanent scientific facts. These are the standing scientific capital of the arc -- the results that survive the substrate's negative verdict and constitute the framework's carry-forward to the next substrate. Each fact is a falsified or characterized claim that the framework now treats as a structural constraint on any future substrate investigation.

9.1 Fact 1: Falsified Collapse-Ordering

Claim falsified. The picture's collapse-ordering hypothesis held that the channels of relational information collapse in a specific order under strain, with the existence channel surviving longest (the black-hole picture's "last channel to go" intuition). The ordering was hypothesized as: string -> direction -> distance -> existence.

Why falsified. Delta's burden dial trajectory shows all channels decaying together, with no clean ordering. String decays fastest (-0.444 -> -0.011 by h=1.3, 97% loss); direction decays at the same rate as distance; existence decays more slowly but monotonically (0.667 -> 0.153 by h=1.3). The hypothesized ordering does not hold; the channels co-decay.

Standing constraint. Any future substrate investigation must NOT assume a clean channel-ordering collapse. Channels must be measured independently; their decay trajectories are substrate-specific, not framework-predictable.

9.2 Fact 2: Falsified Role-Transfer

Claim falsified. The role-transfer hypothesis held that when one channel fails, another channel can take up its role (the black-hole picture's "the boundary carries what the bulk lost" intuition). The hypothesized transfer was: bulk string -> boundary string -> boundary edge-mode when bulk cloaks.

Why falsified. Option A tested this directly. At h=1.3, where bulk composites are at ~0.13 (well below the 0.05 survival threshold), edge composites are at ~0.12 (also below threshold). The edge does not carry the bulk's lost information; the edge composites co-decay with the bulk composites. There is no role-transfer.

Standing constraint. Any future substrate investigation must NOT assume role-transfer without explicit measurement. The edge and bulk channels must be measured independently and their co-decay or independence established empirically.

9.3 Fact 3: Conformal 1/n Floor Artifact

Artifact characterized. The critical-point L_0 floor -- the apparent gap minimum at the substrate's critical point -- is a 1/n finite-N artifact of conformal critical scaling, not a substrate physics feature. The eta two-size test (N=10 vs N=12 under V_strain) gives ratio 0.8379, matching the 1/n prediction 0.8333 to distance 0.005 (essentially exact). The 1/n^2 prediction 0.6944 is at distance 0.14 (too far); the stable prediction 1.0 is at distance 0.16 (too far).

Why this matters. A finite-N critical point's gap scales as v*pi/N (conformal gap formula). Treating this finite-N regularization as a substrate floor would falsely credit the substrate with an L_0 floor it does not possess. The picture's "L_0 floor prevents singularity" would then be a category error (Type-A identity confused with Type-B rate, exactly the conflation Protocol #20 forbids).

Standing constraint. Any future substrate investigation must run a two-size test (or larger) at any claimed L_0 floor. The 1/n scaling class is the artifact signature; 1/n^2 or stable scaling would be substrate physics. The two-size test is mandatory before any R4 (L_0 floor) claim is drawn.

9.4 Fact 4: Multiplet Artifact (Lanczos Blindness)

Artifact characterized. The Lanczos algorithm sees degenerate multiplets as a single eigenvalue cluster and reports the within-multiplet splitting as the "gap." This strikes as a gap=0 reading when the true gap to the next multiplet is small but nonzero (the zeta gap=0 at J=0.3 is the within-triplet splitting; the true singlet gap is 0.001231).

Why this matters. Treating the within-multiplet splitting as the gap would falsely conclude that the substrate is gapless (R4 violation) when it is merely SU(2)-symmetric. The zeta verdict "R4 fails under V_FM" was initially drawn from this artifact; eta Part 1's multiplet diagnostic resolved it, but the artifact had already produced a false R4-strike in the interim.

Standing constraint. Any future substrate investigation must run a multiplet diagnostic from the first data point. The diagnostic computes <psi_i | S_total^2 | psi_i> for each low eigenvalue to identify the SU(2) sector; gap_corrected = E(lowest non-ground-multiplet state) - E(ground). Only gap_corrected may be used in R4 claims.

SECTION 9 VERDICT: Four permanent facts REGISTERED as standing scientific capital. Each is a falsified or characterized claim that any future substrate investigation must respect. The four facts together constitute the framework's diagnostic-constraint set for substrate investigations.

10. The Artifact Taxonomy (Diagnostic Checklist)

The AKLT arc produced a taxonomy of instrument artifacts -- behaviors of the substrate's measurement apparatus that, when read naively, produce false conclusions. Each artifact is characterized by its signature (how to detect it) and its diagnostic (how to resolve it). The taxonomy is registered as a standing checklist that any future substrate investigation must consult before drawing conclusions.

10.1 Artifact 1: Multiplet Blindness

Signature. The Lanczos algorithm returns degenerate eigenvalue clusters as a single value, with within-cluster splitting reported as the gap. Strikes as a gap=0 reading when the true gap to the next multiplet is small but nonzero. Appears in: zeta's V_FM gap=0 at J=0.3 (within-triplet splitting); Option A's h=0 missing 4th degenerate member (eigsh returned 3-fold + first bulk excitation).

Diagnostic. Compute <psi_i | S_total^2 | psi_i> for each low eigenvalue to identify the SU(2) sector (S(S+1) ~ 0 singlet, 2 triplet, 6 quintet, ...). gap_corrected = E(lowest non-ground-multiplet state) - E(ground). Only gap_corrected may be used in R4 claims. The diagnostic must be run from the FIRST data point, not after a gap=0 reading appears.

10.2 Artifact 2: Conformal Floor

Signature. The gap minimum at a critical point scales as v*pi/N (conformal gap formula), giving 1/N scaling with system size. Strikes as an apparent L_0 floor when read at a single N. Appears in: delta's V_strain gap_min 0.0732 at h_c (initially read as a substrate L_0 floor; eta Part 2 showed it is 1/N finite-N artifact).

Diagnostic. Run a two-size test (at minimum) at any claimed L_0 floor. The scaling class is the verdict: 1/N -> conformal artifact (strike); 1/n^2 -> substrate physics (confirmed); stable -> substrate physics with even smaller finite-size corrections (e.g. AKLT exact solvability gives stable at h=0). The two-size test is mandatory before any R4 (L_0 floor) claim.

10.3 Artifact 3: C-Order Indexing

Signature. The C-order (column-major vs row-major) of operator construction can produce index permutation errors that manifest as off-by-one site assignments in correlation functions. Strikes as a distance / direction measurement that is correct in magnitude but wrong in site assignment. Appears in: delta early data (caught by self-audit before verdict draw).

Diagnostic. Verify operator construction against a known analytic limit (e.g. the AKLT vacuum expectation <vac | S^z_0 S^z_1 | vac> = -4/9 should reproduce to machine precision). Cross-check with an independent construction path (vectorized vs. Kronecker-chain).

10.4 Artifact 4: E_joint Column

Signature. The joint product (distance * direction) can show apparent enhancement at critical points that is actually a cancellation of leading-order decay, not robustness-by-protection. Strikes as a "rescue" of the position channel when read without distinguishing robustness-by-anti-correlation from protection. Appears in: epsilon's ell=2 doubling at h_c.

Diagnostic. Test the complementarity mechanism at multiple ell values; if the enhancement is bound to a specific ell regime where channels anti-correlate, it is robustness-by-anti-correlation (not protection). If the enhancement is uniform across ell, it is genuine protection.

10.5 Artifact 5: Lanczos Degenerate-Manifold (related to Artifact 1)

Signature. Distinct from multiplet blindness (which is about cluster-splitting misread as gap), this artifact is about eigsh returning arbitrary orthogonal vectors within a degenerate subspace, mixing S^z sectors. Strikes as non-integer S^z_tot values for eigenstates. Appears in: Option A's h=0 lowest 4 eigenvalues with S^z_tot values {+0.47, -0.13, -0.20, -0.01} (mix of SU(2) sectors).

Diagnostic. Same multiplet diagnostic as Artifact 1. The non-integer S^z_tot is the signature of subspace mixing; the diagnostic resolves the SU(2) sector of each cluster rather than each individual eigenstate.

10.6 Artifact Taxonomy as Diagnostic Checklist

#	Artifact	Signature	Diagnostic
1	Multiplet blindness	Gap=0 at SU(2)-symmetric point	S_total^2 multiplet diagnostic from first data point
2	Conformal floor	Gap_min scales as 1/N	Two-size test; ratio N12/N10 ~ 0.83 = artifact
3	C-order indexing	Off-by-one site assignments	Verify against AKLT vacuum analytic limit
4	E_joint column	Apparent enhancement at h_c	Multi-ell test; uniform = protection, ell-bound = anti-correlation
5	Lanczos degenerate-manifold	Non-integer S^z_tot values	Same multiplet diagnostic; resolve SU(2) sector per cluster

SECTION 10 VERDICT: Artifact taxonomy REGISTERED as standing diagnostic checklist. Any future substrate investigation MUST consult this checklist before drawing conclusions. The five artifacts together constitute the framework's instrument-behavior knowledge capital.

11. Updated Certified Master Hierarchy

The certified master hierarchy is preserved unchanged from v6.5-RF8 (with the audit-patch corrections from v6.1-RF1 applied). The hierarchy is the verified-number spine on which the framework's gap statements rest; it is the framework's stable bedrock, unaffected by the AKLT arc's negative verdict on the projection picture.

Delta_DL = 1/6 [Type-A, algebraic, VN-18]
<= Delta_Knabe^(4) = 0.173434 [Type-A, exact diagonalization, VN-10]
<= Delta_Knabe^(6) = 0.248064 [Type-A, exact diagonalization, VN-11]
<= Delta_Knabe^(8) = 0.275907 [Type-A, exact diagonalization]
<= Delta_Knabe^(10) = 0.291075 [Type-A, exact diagonalization]
<= Delta_infinity = 0.350115 [Type-B, -> 3^{-L} periodic exponential, VN-14/15]
<= epsilon_6 = 0.398451 [Type-A, exact diag, VN-09]
<= epsilon_4 = 0.448956 [Type-A, irreducible quartic, VN-08]
<= epsilon_3 = 0.500000 [Type-A, spin-1 algebra, VN-04]

Plus the structural layer (preserved from v6.5-RF8):

g_FS = (1/2) Hess(Phi) [Type-A, Kahler geometry, VN-19]
=> B_geom well-posed on F_MM = 0 [Type-A, VN-20]
=> Hierarchical RP (parabolic flow) [Type-B, mechanism-determined rate]
=> S5 discharged (primitive count 5 -> 4)
=> RF-4 internalized [Type-A layerability + Type-B contraction]

11.1 Hierarchy Status Tags (per Protocol #20)

Quantity	Value	Provenance	Status	Type tag
Delta_DL	1/6	C*-algebra B*B identity (RF-1)	[Established]	[=]
Delta_Knabe^(n)	0.173, 0.248, ...	Knabe finite-block theorem	[Established]	[=]
Delta_infinity	0.350115	Periodic exponential extrapolant	[Established]	[->: 3^{-L}]
epsilon_n	0.398, 0.449, 0.500	Exact diagonalization	[Established]	[=]
g_FS = (1/2) Hess(Phi)	1/2 (exact)	Kahler geometry	[Established]	[=], VN-19]
F_MM = 0	0 (constraint)	Moment map	[Established]	[=], VN-20]
B_geom >= 0	0 (lower bound)	Variance non-negativity	[Established]	[=]
gamma_DL	1/sqrt(3)	RF-1 sandwich norm	[Established]	[->: rate], VN-17]

12. Open Hypothesis: R4' Floor-Under-Strain

The user's R4' floor-under-strain hypothesis remains open. Distinct from R4 (the critical-point L_0 floor, struck in eta as 1/N finite-N artifact), R4' is the hypothesis that a ground-state L_0 floor survives in the extreme-strain regime (h >> h_c), as a substrate feature. Eta-extreme confirmed the user's distinction: the extreme-strain floor IS a substrate feature (stable with N across h >> h_c, gap -> h as h -> inf). But the floor does not prevent singularity in the original R4 sense -- the singularity (gap=0) is at the critical point h_c, and the extreme-strain regime is in a DIFFERENT GAPPED PHASE (large-D) with finite gap by construction. The R4' hypothesis lives on the next substrate.

12.1 R4' Status (post-arc, open)

Distinction (CONFIRMED, eta-extreme). R4' floor-under-strain is a real substrate feature in the extreme-strain regime, distinct from the critical-point L_0 floor. The two are not the same object; the former survives the thermodynamic limit, the latter does not.

Tension (CONFIRMED, eta-extreme). R4' floor-under-strain and R3 survival are mutually exclusive on the AKLT substrate: the extreme-strain regime (where R4' is restored) is the all-m=0 product state (where R3 collapses to zero). This is the SAME R3/R4 tension the auditor identified, persisting across both the critical-point and extreme-strain regimes.

Open question (carries to next substrate). Does the R4' floor-under-strain survive on a substrate that satisfies the primitive-vs-emergent criterion? Such a substrate would have boundary observables that do not co-fold with bulk observables under strain; the R4' floor might then preserve relational structure (R3) in the extreme-strain regime, breaking the tension. The next substrate investigation must test this.

12.2 R4' Versus R4 (Distinction Table)

Aspect	R4 (L_0 floor)	R4' (floor-under-strain)
Location	Critical point h_c	Extreme-strain regime h >> h_c
Scaling	1/N (conformal)	Stable with N (substrate)
Status (AKLT)	STRUCK (eta two-size test)	RESTORED (eta-extreme confirmed)
Relation to singularity	Alleged to prevent; does not (1/N artifact)	Does not prevent; singularity is at h_c, not extreme
Relation to R3	R3 partial at h_c (sum ratio 0.383)	R3 collapses in extreme strain (-> 0)
Open on next substrate	Struck verdict carries; floor must be tested with two-size protocol	Hypothesis open; substrate with primitive-vs-emergent admissibility may break the R3/R4' tension

SECTION 12 VERDICT: R4' floor-under-strain is [OPEN -- Hypothesis carries to next substrate]. The distinction from R4 (critical-point L_0 floor, struck) is established. The R3/R4' tension is confirmed persistent on AKLT; whether the tension breaks on a primitive-vs-emergent-admissible substrate is the open question the next substrate investigation must address.

13. Standing Debts Register

The framework carries a small set of standing debts -- items that have been deferred across multiple increments and remain open. Each is logged with its source, its carry-forward path, and its priority. The register is preserved across the v7.0 rewrite; no debt is closed or added by this document.

ID	Debt	Source	Priority	Status / Carry-forward
VN-62	Plaquette Lie holonomy script (T-7 followup)	WP-4 (T-7 Galactic Dynamics)	Medium	Deferred to next substrate investigation; plaquette-ring construction needs rewrite for non-AKLT substrates
Constraint-set spectra	19th round of constraint set spectra	RF-2 audit cycle	Low	Carries forward; not blocking; would close analytically rather than numerically if achievable
T_3 PSD	Positive-semi-definiteness of T_3 (3-site transfer operator)	RF-4 internalization (v6.5-RF8)	Medium	Carries forward; PSD status would close the detectability contraction rate at the analytical level; currently Type-B unspecified
T-1 layer conditions	Layer-condition specification for T-1 (Lorentzian signature)	WP-5C audit (v6.25)	Medium	OPEN. The Lorentzian signature emergence (T-1) remains algebraically underived; the minus sign is stipulated by definition, not derived from algebraic structure (Protocol #21 violation, pending closure)

13.1 Debt Priority Interpretation

Medium priority debts (VN-62, T_3 PSD, T-1 layer conditions) are not blocking but would close structural questions at the analytical level if achieved. They carry forward to the next substrate investigation, where some may resolve naturally (e.g. T_3 PSD on a substrate with a simpler transfer-operator spectrum).

Low priority debts (constraint-set spectra, 19th round) are not blocking and would benefit from analytical rather than numerical closure if achievable. They carry forward indefinitely; closure is not required for framework progress.

14. Status of Research Frontiers

The framework's research-frontier register is updated to reflect the post-arc state. RF-1 through RF-8 are closed (preserved from v6.5-RF8). RF-9 (the AKLT substrate investigation) is closed with a negative verdict -- the substrate's question is answered, and the negative answer is permanent scientific capital. RF-10 (substrate-change fork) is launched.

Frontier	Epistemic class	Status	Formal resolution
RF-1 (DL Sandwich Norm)	Type-A [=]	[CLOSED -- Established]	gamma_DL = 1/sqrt(3) via RF-1 sandwich construction
RF-2A (Boundary Mechanism)	Type-A [=]	[CLOSED -- Established]	Ferromagnetic effective Hamiltonian
RF-2B (Two-Scale Asymptotic Gap)	Type-B [->: 1/n^2 bulk, 1/n^3 boundary]	[CLOSED -- Established]	Bulk kinetic + boundary exchange
RF-3 (epsilon_4 Quartic Root)	Type-A [=]	[CLOSED -- Established]	Irreducible quartic
RF-4 (Detectability Lemma)	Type-A layerability + Type-B contraction	[CLOSED -- Internalized]	C-02 = Type-B manifestation of Type-A layerability
RF-7 (Boundary-Modified Knabe)	Type-B [->: 1/n]	[DISSOLVED -- Redundant]	Standard Knabe converges at O(1/n)
RF-8 (Hierarchical RP / Hessian)	Type-A [=] foundation + Type-B parabolic flow	[CLOSED -- Established]	Hessian identity + parabolic flow
RF-9 (AKLT Substrate Investigation)	Type-B [->: substrate-specific verdict]	**[CLOSED -- Negative verdict, permanent capital]**	Six-round arc; picture falsified in every dynamical form on AKLT; scoping theorem (section 8) applied; four permanent facts + artifact taxonomy registered; substrate-change fork (RF-10) launched
RF-10 (Substrate-Change Fork)	Type-B [->: open]	**[OPEN -- Launched]**	Appendix A: four candidate substrates, selection criterion (primitive-vs-emergent + kinematic-boundary), recommended order

14.1 Open Frontier Summary

RF-10 (Substrate-Change Fork) is the only open frontier. Its goal is to identify and test a substrate that satisfies the primitive-vs-emergent admissibility criterion (section 3.4) and supplies a kinematic boundary-bulk decomposition. The four candidate substrates in Appendix A are evaluated against this criterion; the recommended order prioritizes a control substrate (Ising TF, null hypothesis) followed by a decisive substrate (2D SPT, kinematic boundary protection).

SECTION 14 VERDICT: RF-9 CLOSED with negative verdict (permanent capital). RF-10 LAUNCHED. RF-1 through RF-8 unchanged (closed). The framework's open frontier set is {RF-10} only.

15. Summary

This document is the post-arc rewrite of the RCF framework. It subsumes v6.5-Protocol#20 (Type-Tagging of Vanishing Claims), v6.5-RF8 (Hierarchical RP / Hessian Minimization), v6.1-RF1 audit patches, and the six-round AKLT substrate arc (delta, epsilon, zeta, eta, eta-extreme, Option A) into a single coherent specification.

The framework's structural commitments are preserved unchanged. The Hessian identity g_FS = (1/2) Hess(Phi) remains the foundational Type-A tautology on which the parabolic gradient flow is built. The Type-A / Type-B classification remains the mandatory vocabulary for every vanishing claim. The certified master hierarchy remains the verified-number spine. The protocol system (#15-#20) remains the standing audit apparatus. The primitive count remains four (S5 discharged in v6.5-RF8).

What changes is the picture's scope. The projection picture survives ONLY as the standard-GR cloaking asymmetry (kinematic, phase-independent). It is falsified in every dynamical form the AKLT substrate could host. The arc's negative verdict is permanent scientific capital: four permanent facts (falsified collapse-ordering, falsified role-transfer, conformal floor artifact, multiplet artifact), a complete map of the substrate's trade-off surface (gap vs. information, monotone, no interior extremum), a measured endpoint (h -> infinity, all channels cloak together), and a diagnostic taxonomy of the artifacts the substrate's instruments produce when their outputs are read naively.

Most importantly, the arc surfaced a foundational correction that this document absorbs as a structural axiom: in RCF, every physical observable is emergent, not primitive. The AKLT substrate -- like nearly every spin-chain substrate -- is built with primitive observables (spin operators and their correlations). The arc treated them as primitive because that is what the substrate hands you. But RCF's foundational commitment is that physical observables are always emergent from the relational coarse-graining, never primitives of the framework. The mismatch is the structural source of the negative verdict on every dynamical form the AKLT substrate could host.

The framework's open frontier is RF-10 (Substrate-Change Fork). The four candidate substrates (Ising TF control, S=1/2 Heisenberg critical, higher-S SPT richer, 2D SPT topological) are evaluated against the primitive-vs-emergent admissibility criterion in Appendix A. The recommended order prioritizes a control substrate (Ising TF) to establish the null hypothesis, followed by a decisive substrate (2D SPT) whose boundary protection is kinematic rather than SPT-dynamical. The R4' floor-under-strain hypothesis (section 12) carries forward as an open question: does the floor survive on a substrate that satisfies the primitive-vs-emergent criterion, and does it preserve relational structure (R3) in the extreme-strain regime, breaking the AKLT-substrate tension?

The framework is complete in the sense that no further Type-A primitives are required. The remaining work is Type-B: substrate-specific investigations on substrates selected by the primitive-vs-emergent criterion, with the four permanent facts and the artifact taxonomy as standing diagnostic capital. The AKLT laboratory has answered its final question in the negative for every location it possesses; the conversation moves to the next substrate.

Appendix A. Substrate-Change Fork (Forward-Looking Roadmap)

This appendix operationalizes the substrate-change fork. Four candidate substrates are evaluated against the substrate-evaluation criterion (section 3.4 / section 8.4). The criterion has three conditions: (i) primitive-emergent admissibility (every primitive in A_prim(S) expressible as a function of C_L-class observables); (ii) nontrivial boundary-bulk decomposition (boundary algebra distinct from bulk algebra); (iii) kinematic (phase-independent) rather than SPT-dynamical (phase-dependent) boundary-bulk decomposition.

A.1 The Four Candidate Substrates

#	Substrate	Primitive-emergent	Boundary-bulk	Kinematic	Role
(a)	Ising transverse-field chain (1D)	Small gap (sigma_x vs sigma_z prim)	Trivial (no edge modes)	N/A (no boundary)	CONTROL (null hypothesis)
(b)	S=1/2 Heisenberg chain (1D, critical)	SU(2)-symmetric prim algebra; CFT-dual emergent interpretation	Critical CFT has boundary-bulk via conformal boundary states	YES (kinematic, CFT structure)	Critical substrate, CFT-dual
(c)	Higher-S SPT chain (S >= 2 AKLT)	S=2+ spin operators; richer edge multiplet structure	Yes (SPT edge modes; richer than S=1)	NO (SPT-dynamical, phase-dependent)	Richer protection channels
(d)	2D SPT (e.g. Levin-Gu model)	Plaquette operators; topologically encoded emergents	Yes (topological boundary modes; robust by topology)	YES (topological = kinematic)	DECISIVE: kinematic boundary protection

A.2 Selection Criterion Applied

AKLT (baseline, struck substrate). Satisfies (i) marginally (spin operators admit coarse-graining interpretations via MPS but the gap is large); fails (ii) (SPT-edge observables are boundary-local versions of bulk observables, not emergents of a distinct boundary algebra); fails (iii) (SPT-edge protection is phase-dependent, h ~ 1 destroys it).

(a) Ising transverse-field. Satisfies (i) cleanly (sigma_x, sigma_z admit Jordan-Wigner emergent interpretation). Fails (ii) (1D Ising has trivial boundary; no edge modes). Fails (iii) by failure of (ii). **Role:** CONTROL. The Ising TF chain is a null-hypothesis substrate: it has well-understood phase structure, exact JW transformation, and known conformal-critical behavior at h_c = 1. The framework runs the same arc instruments (delta/epsilon/eta protocols) on it to establish the null-hypothesis trajectories: what the instruments measure when the substrate is known to have no boundary-bulk structure. This calibrates the instruments themselves.

(b) S=1/2 Heisenberg chain. Satisfies (i) cleanly (spin-1/2 operators admit a complete CFT-dual emergent interpretation; the Heisenberg chain is dual to the SU(2)_1 WZW CFT). Satisfies (ii) (the CFT has boundary-bulk structure via conformal boundary states; the boundary algebra is the chiral algebra of the CFT, distinct from the bulk algebra). Satisfies (iii) (CFT boundary-bulk is kinematic, defined by the CFT structure, not by a phase). **Role:** CRITICAL SUBSTRATE. The Heisenberg chain is gapless, so the R4 (L_0 floor) question is replaced by the question of whether the CFT central charge c=1 and the conformal boundary states carry the relational information that survives strain.

(c) Higher-S SPT chain (S >= 2). Same primitive-emergent profile as AKLT (large gap), but richer edge multiplet structure (S=2 AKLT has edge spin-1 multiplet, vs S=1 AKLT's edge spin-1/2). Satisfies (i) marginally; satisfies (ii) more richly (richer boundary algebra); fails (iii) (still SPT-dynamical, phase-dependent). **Role:** RICHER PROTECTION CHANNELS. The S=2 AKLT substrate tests whether richer SPT-edge structure survives V_strain. If S=2 SPT-edge survives where S=1 did not, the framework learns that the SPT-dynamical boundary protection is richer than S=1 exposed; if S=2 also fails, the failure is structural to SPT-dynamical protection itself.

(d) 2D SPT (Levin-Gu model). Satisfies (i) cleanly (plaquette operators admit topologically encoded emergent interpretation; the Levin-Gu model's ground state has topological order with emergent anyonic excitations). Satisfies (ii) cleanly (topological boundary modes are robust by topology; boundary algebra is the anyon algebra, distinct from bulk). Satisfies (iii) cleanly (topological = kinematic; the boundary protection is built into the substrate's topology, not into a phase). **Role:** DECISIVE. The 2D SPT substrate is the cleanest test of the primitive-vs-emergent criterion: if the projection picture survives here, the criterion is validated; if it fails here, the criterion needs revision.

A.3 Recommended Order

Phase 1: Control (Ising TF, candidate (a)). Establish the null-hypothesis instrument trajectories. Run the delta / epsilon / eta protocols on the Ising TF chain across the h=0 to h > h_c range. The expected outcome: the instruments measure smooth decay across the critical point with no boundary protection (1D Ising has no boundary). This calibrates the instruments and provides the baseline against which the decisive substrates' departures are measured.

Phase 2: Decisive (2D SPT, candidate (d)). Run the same protocols on the Levin-Gu 2D SPT model. The expected outcome: if the primitive-vs-emergent criterion is correct, the 2D SPT substrate should exhibit boundary observables that do NOT co-fold with bulk observables under strain. The topological boundary modes should retain their structure even when the bulk is strained, because they are kinematically protected by topology rather than dynamically protected by a phase.

Phase 3 (conditional): S=2 SPT (candidate (c)). Only if Phase 2 fails -- to test whether richer SPT-edge structure survives where 2D topology did not. If Phase 2 succeeds, Phase 3 is unnecessary; if Phase 2 fails, Phase 3 tests whether the failure is structural to SPT-dynamical protection (S=2 also fails) or specific to 2D topology (S=2 succeeds).

Phase 4 (conditional): S=1/2 Heisenberg (candidate (b)). Only if both Phase 2 and Phase 3 fail -- to test whether CFT-dual substrates with conformal boundary states host the projection picture. The Heisenberg chain is gapless, so the R4 question is replaced by the central-charge question; this is a substantially different test and is deferred to last.

A.4 Standing Constraints (carried from §9)

All four substrates must be investigated with the four permanent facts (section 9) and the artifact taxonomy (section 10) as standing diagnostic capital. Specifically: (i) no clean channel-ordering collapse may be assumed; (ii) no role-transfer may be assumed without explicit measurement; (iii) any claimed L_0 floor must pass a two-size test before being accepted as substrate physics; (iv) a multiplet diagnostic must be run from the first data point; (v) the artifact taxonomy must be consulted before any verdict is drawn.

APPENDIX A VERDICT: Substrate-change fork LAUNCHED. Four candidate substrates evaluated against the primitive-vs-emergent admissibility criterion. Recommended order: Phase 1 (Ising TF control) -> Phase 2 (2D SPT decisive) -> Phase 3 (S=2 SPT, conditional) -> Phase 4 (S=1/2 Heisenberg, conditional). Standing

constraints from §9 enforced.

A.5 Closing Note

The AKLT arc closed a chapter of the framework. The substrate-change fork opens the next. The framework's structural commitments are unchanged; what changes is which substrate the framework asks its question on. The primitive-vs-emergent axiom (section 3) is the structural correction the arc surfaced; the substrate-evaluation criterion (section 3.4 / Appendix A) is its operational consequence. The next substrate investigation begins with the Ising TF control (Phase 1) and proceeds to the 2D SPT decisive test (Phase 2). The R4' floor-under-strain hypothesis (section 12) is the open question the decisive test must address.

Appendix B. Glossary of Key Terms

This glossary defines the framework's specialized vocabulary. Terms are listed alphabetically. Where a term has both a general meaning and a framework-specific meaning, the framework-specific meaning is given.

Term	Definition
AKLT substrate	The spin-1 Affleck-Kennedy-Lieb-Tasaki chain, H_AKLT = Sigma [1/2 S_j.S_{j+1} + 1/6 (S_j.S_{j+1})^2]. The substrate of the six-round arc (section 7).
Artifact taxonomy	The standing diagnostic checklist of instrument-behavior artifacts (section 10): multiplet blindness, conformal floor, C-order indexing, E_joint column, Lanczos degenerate-manifold.
Burden dial	The strain-burden parameter (h or J) that the framework dials across a substrate to test the projection picture. Two burdens in the arc: V_strain (existence channel), V_FM (temporal channel).
Certified master hierarchy	The verified-number spine: Delta_DL <= Delta_Knabe^(n) <= Delta_infinity <= epsilon_n (section 11). Preserved from v6.5-RF8.
Conformal floor artifact	The 1/N scaling of a critical-point gap minimum, characterized by the eta two-size test (Fact 3, section 9.3). Distinguished from a true substrate floor by the two-size protocol.
Emergent observable	An observable that is a function of C_L-equivalence class observables, where C_L is a relational coarse-graining. Per the primitive-vs-emergent axiom (S5', section 3), every physical observable in RCF is emergent, never primitive.
Existence channel	The observable family <(S^z)^2> and its burden V_strain = +h sum (S^z)^2. Drives the Haldane -> large-D transition in the AKLT substrate.
F_MM (moment map)	The moment-map constraint F_MM(psi) = (H psi - E_0 psi) / ||psi||^2 (mod ground-state projection). F_MM = 0 defines the constraint manifold M_constraint. Type-A [=] by symmetry; registered as VN-20.
g_FS (Fubini-Study metric)	The Kahler metric on CP(H). Equals (1/2) Hess(Phi) per the Hessian identity (Theorem 3.1, VN-19).
Geometric burden (B_geom)	B_geom(psi) = Var(g_FS)|_psi. Bounded below by 0 (variance). The functional whose gradient generates the parabolic flow (Hierarchical RP).
Hessian identity	g_FS = (1/2) Hess(Phi), where Phi = 2 K_FS = 2 log <psi|psi>. Type-A [=] by Kahler geometry; foundational tautology of the framework.
Hierarchical RP	Hierarchical Robustness Principle: the parabolic layer-sequential flow on (M_constraint, g_FS), decomposing into Type-A constraint satisfaction (Layer 1) and Type-B metric relaxation (Layer 2).
L_0 floor	The minimum gap value reached by a substrate under burden dial. In the AKLT arc, the critical-point L_0 floor (delta's 0.0732) was struck as 1/N conformal artifact (eta two-size test). Distinct from R4' floor-under-strain.
Lanczos manifold blindness	The artifact where Lanczos returns within-multiplet splitting as the gap (Fact 4, section 9.4). Resolved by the multiplet diagnostic (S_total^2 sector identification).
M_constraint	The constraint manifold {psi in CP(H) : F_MM(psi) = 0}, on which the parabolic flow is defined. Type-A [=] by moment-map non-degeneracy.
Multiplet diagnostic	The protocol of computing <psi_i | S_total^2 | psi_i> for each low eigenvalue to identify the SU(2) sector, then computing gap_corrected = E(lowest non-ground-multiplet state) - E(ground). Mandatory from the first data point.
Picture (the projection picture)	The framework's hypothesis that relational information survives strain via a bulk-to-boundary projection mechanism. Scoped to its surviving kinematic form (standard-GR cloaking asymmetry) by the boundary statement (section 8).

Term	Definition
Primitive observable	An observable defined directly in terms of substrate operators (e.g. spin operators S^z, S^+, S^-) without reference to a coarse-graining equivalence class. The AKLT substrate's algebra is primitive; RCF requires emergent.
Primitive-vs-emergent axiom (S5')	The structural axiom (section 3) that every physical observable in RCF is emergent from the relational coarse-graining, never primitive. Surfaced by the AKLT arc; absorbed as foundational in v7.0.
Protocol #15-#20	The standing audit apparatus: #15 Independent Recomputation, #16 Variational Principle Enforcement, #17 Parity & Extremum Derivative, #18 Numerical Equivalence Machine-Check, #19 Bibliographic External Lock, #20 Type-Tagging of Vanishing Claims.
R1, R2, R3, R4 (picture's claims)	The four claims the projection picture makes about substrate behavior under strain: R1 (existence HIDDEN, not COLLAPSED); R2 (symmetric cloaking -- RETRACTED as asymmetric + distortion); R3 (position+scale survive); R4 (L_0 floor). R4' is the floor-under-strain variant.
RF-9, RF-10	Research Frontiers 9 (AKLT substrate investigation, CLOSED with negative verdict) and 10 (Substrate-Change Fork, OPEN -- the only open frontier post-arc).
S1-S4 (primitive postulates)	The four primitive postulates of the framework: S1 Hilbert space, S2 Hamiltonian spectrum, S3 symmetry group and moment map, S4 variational principle. S5 was discharged in v6.5-RF8.
Scoping theorem	The picture's boundary statement (section 8): the projection picture survives ONLY as the standard-GR cloaking asymmetry (kinematic, phase-independent); falsified in every dynamical form the AKLT substrate could host.
SPT-edge protection	The boundary protection provided by symmetry-protected topological phases. In the AKLT substrate, exists at h=0 (edge = 2 x bulk) but fragile (destroyed by h ~ 1). Falsified for the AKLT substrate by Option A.
Substrate-evaluation criterion	The operational consequence of the primitive-vs-emergent axiom (section 3.4 / section 8.4 / Appendix A). Three conditions: (i) primitive-emergent admissibility; (ii) nontrivial boundary-bulk decomposition; (iii) kinematic rather than SPT-dynamical.
Temporal channel	The observable family <S_j.S_{j+1}> and its burden V_FM = -J sum S_j.S_{j+1}. Drives the Haldane -> Ising-FM transition in the AKLT substrate.
Two-size test	The protocol of running a burden scan at two system sizes (typically N=10, N=12) and computing the ratio. Ratio ~ 0.83 = 1/N conformal artifact; ~ 0.69 = 1/n^2 substrate physics; ~ 1.0 = stable substrate feature.
Type-A [=]	Vanishing-claim tag for exact identity forced a priori by symmetry, algebra, or representation theory. Constrains the FORM of related quantities. Per Protocol #20 (section 4).
Type-B [->: rate]	Vanishing-claim tag for asymptotic limit whose rate O(n^-p) is determined a posteriori by physical mechanism. Constrains only the existence of convergence. Per Protocol #20 (section 4).
VN (Verified Number)	An entry in the framework's verified-number register. Each VN has a value, a provenance class (E/IV/C/F/O/L), a status tag ([Established], etc.), and a Type tag ([=] or [->: rate]).
V_strain, V_FM	The two strain burdens of the AKLT arc. V_strain = +h sum (S^z)^2 (existence channel, drives Haldane -> large-D); V_FM = -J sum S.S (temporal channel, drives Haldane -> Ising-FM).

B.1 Symbol Quick Reference

Symbol	Meaning
Delta	Bulk gap E_1 - E_0 (singlet-channel corrected)
Delta_infinity	Thermodynamic-limit gap (0.350115)
Delta_DL	Detectability-lemma gap (1/6)
Delta_Knabe^(n)	Knabe finite-block bound at size n

Symbol	Meaning
epsilon_n	Local AKLT gap at chain length n
g_FS	Fubini-Study metric on CP(H)
Phi	Kahler potential 2 log <psi|psi>
F_MM	Moment-map constraint (algebraic compatibility)
M_constraint	Constraint manifold {F_MM = 0}
B_geom	Geometric burden Var(g_FS)
gamma_DL	Detectability-lemma contraction (1/sqrt(3))
h, J	Strain burden parameters (V_strain, V_FM resp.)
h_c	Critical point of V_strain dial (~0.9-0.95)
N	System size (typically 10 or 12)
S^z, S^+, S^-	Spin-1 operators
S_total^2	Total-spin-squared operator (multiplet diagnostic)
O_string(i,j)	String order parameter prod S^z_k (i ≤ k ≤ j)
[=]	Type-A vanishing-claim tag (exact identity)
[->: rate]	Type-B vanishing-claim tag (asymptotic limit, mechanism-determined rate)